

North Coast (Areas 3 & 4) Recreational Angling Creel Survey 2018

FINAL REPORT



Prepared for:

North Coast - Skeena First Nations Stewardship Society
Prince Rupert, BC

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EXECUTIVE SUMMARY

Given its potential impact on multiple transboundary stocks of salmon and groundfish, the North Coast (Areas 3 and 4) recreational fishery must be monitored. Effort and harvest data are required for management of fisheries and monitoring of indicator stocks. Since 1997, creel surveys have been used to estimate effort and harvests, with the added benefit of the collection of biological samples such as scales, and Coded Wire Tags. The North Coast Skeena First Nations Stewardship Society (NCSFNSS) has managed this program since 2015, with support from the Pacific Salmon Commission Northern Fund. The survey consisted of access point interviews and aerial overflights, for the months of June, July and August 2018. Lodge-based recreational harvest and effort were monitored by DFO. Survey design and implementation, and analytical methods were similar to and consistent with those from the past several fishing seasons.

From 1 June to 31 August 2018, a total of 1,482 vessel interviews (involving 4,410 anglers) were collected by staff stationed at one of three access points (located in Prince Rupert or Port Edward) during 176 survey shifts. 23 aerial effort surveys were conducted. High priority was placed on gathering biosamples and fin-clip incidence rates from Chinook (*Oncorhynchus tshawytscha*) and Coho (*O. kisutch*) salmon, and length measurements from Pacific Halibut (*Hippoglossus stenolepis*). Survey effort was adequate, as relatively good precision was achieved for the harvest estimates for most of the main taxa (all precision goals were met).

Total fishing effort was estimated to be 10,770 vessel trips. In total, an estimated 11,828 Chinook Salmon were caught, with 5,830 fish kept and 5,997 released. It was estimated that 11,030 Coho Salmon were caught, with 10,464 fish kept and 567 released. Total Pacific Halibut catch was estimated to be 19,045 individuals, with 10,853 fish kept and 8,192 released. Biomass of harvested Pacific Halibut was estimated at 62.3 net tonnes (137,249 lb), including 23.3 t (51,504 lb) in Area 3 and 38.9 t (85,843 lb) in Area 4. In terms of round weight, Pacific Halibut harvest was 82.4 t (181,769 lb) including 30.9 t (68,080 lb) in Area 3 and 51.6 t (113,689 lb) in Area 4. Total harvest estimates included 4,270 Lingcod (*Ophiodon elongates*), 7,903 rockfish (*Sebastes* spp.), and 1,187 groundfish other than Pacific Halibut and Lingcod.

Creel surveyors biosampled 285 Chinook Salmon, including length measurements (median 68 cm), adipose fin-clip incidence rates (15% were clipped), and scale samples. For Coho Salmon, 346 fish were biosampled, including the measurement of fork lengths (median 67 cm), and assessment of adipose fin-clip incidence rates (1.6% were clipped). For 37 Chinook and 4 Coho salmon with fin clips, heads were collected for submission to the Salmon Head Recovery Program. Biosampled Pacific Halibut (n=543) had a median length of 82 cm. Average net and round weights were 6.0 kg (13.1 lb) and 7.9 kg (17.4 lb) in Area 3, and 5.6 kg (12.4 lb) and 7.4 kg (16.4 lb) in Area 4, respectively.

Recreational fishing effort, salmon CPE, and salmon catch in Areas 3 & 4 have been increasing over the last 20 years, in some cases dramatically, yet in 2018 Chinook and Coho salmon harvests were at their lowest levels since 1996. Pacific Halibut harvest did not show a strong among-year trend, likely the result of effective management measures that reduced daily bag limits and imposed maximum size limits. So while increasing trends highlight the need for continued monitoring of this fishery into the future, there is evidence that management measures can keep harvest under control.

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INTRODUCTION

The North Coast recreational fishery (in DFO's Pacific Region, Management Areas 3 and 4) harvests multiple species, mainly targeting Chinook (*Oncorhynchus tshawytscha*) and Coho (*O. kisutch*) salmon, but also Pacific Halibut (*Hippoglossus stenolepis*), Lingcod (*Ophiodon elongates*), and other groundfish, rockfish (*Sebastes* spp.) and invertebrate species. Residents and tourists both enjoy sport fishing on the North Coast, and over the past 20 years effort has increased substantially (Robichaud and Addison 2018). In light of increasing pressure, and of conservation concerns for some target stocks, there is a need to understand harvest from the Area 3 and 4 recreational fishery, with respect to catch rates and stock origins. And since harvested salmon can originate from BC or US (enhanced and wild) stocks, there is a critical need for this creel program to determine fishery impacts, especially for stocks of concern.

In addition to producing harvest and effort estimates with known error, this creel program has the added benefit of collecting direct field observations of catch, which allows for species identifications and harvest quantities to be verified, and biological samples to be collected from the recreational fishery. For example, CWT data, which are important components of the annual stock assessment data, and are required to monitor trends and manage the fisheries for Nass and Skeena river Chinook and Coho salmon stocks, come from the heads that our observers collect from randomly sampled adipose-clipped Chinook and Coho salmon. As such, the creel survey program helps the Pacific Salmon Commission develop improved information for resource management, including better stock assessment, data acquisition, and improved scientific understanding of factors that might limit salmon production.

The Area 3 and 4 creel survey program has operated during each recreational fishing season since 2008, estimating fishing effort and species-specific catch rates (Van Tongeren and Winther 2010, Robichaud and Addison 2018). Starting in 2015, the responsibility for carrying out the creel survey shifted from Fisheries and Oceans Canada (Prince Rupert) to the North Coast Skeena First Nations Stewardship Society (NCSFNSS), with support from the Pacific Salmon Commission Northern Fund. NCSFNSS's creel survey followed a design similar to that used in the past by Fisheries and Oceans Canada, which, in turn, was based on an adaptation of the Strait of Georgia Creel Survey design, developed in 1982 by LGL Limited (English et al. 2002).

This report documents the 2018 recreational catch and effort statistics for fishing activities in Pacific Fishery Management Areas 3 and 4, and describes the program activities, data collection methodology and the analytical approaches that were used to generate the estimates.

SCOPE

The geographic scope of the creel survey was the same as in the previous nine years (Van Tongeren and Winther 2010, Robichaud et al. 2016, 2017, Robichaud and Addison 2018), i.e., Management Areas 3 and 4 (from Portland Inlet, through Chatham Sound to Porcher Island; Figure 1). For the purposes of data collection and analysis, the study area was divided into 12 geographic strata ('subareas'), including three subareas in Area 3 (3I-3K) and nine subareas in Area 4 (4A-4H, and 4L).

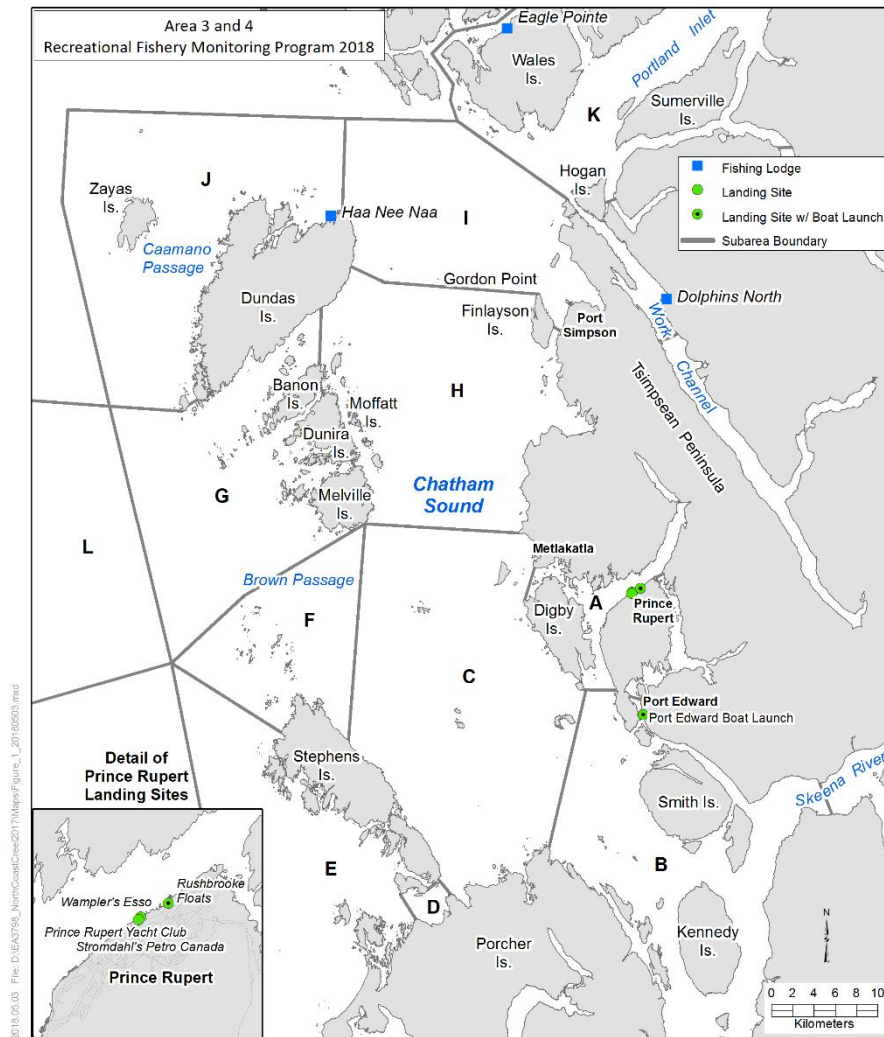


Figure 1. Map of the North Coast study area, showing the major landing sites, fishing lodges, and the boundaries of the subareas, which were used as spatial strata in the creel survey design.

The temporal scope of the full study was from June 1st through August 31st, 2018. Survey schedules were designed to provide sufficient data to derive statistically robust catch and efforts estimates for each month and subarea. However, mid-season it was decided to use four temporal strata, termed “periods”, which better coincided with changes in the management measures regulating the fishery than did the biologically ‘arbitrary’ calendar months. For ease of among-year comparison, the period-specific data were reconstituted into monthly summaries wherever possible. For this reason, the second period was sometimes divided into two sub-periods (Table 1): one sub-period for the June days (June 16 – 30), and the other sub-period for the July days (July 1 – 9).

The study included creel analysis of all major local sportfish species, with greater emphasis placed on Chinook and Coho salmon, Pacific Halibut and Lingcod.

Table 1. Delineations of temporal strata. The four Periods coincided with changes in the fishery regulations. Periods were further divided into five Sub-periods to allow data to be easily grouped into three Months.

<i>Period</i>	<i>Sub-period</i>	<i>Month</i>
June 1 - 15	June 1 - 15	June
June 16 - July 9	June 16 - 30	
	July 1 - 9	July
July 10 - 31	July 10 - 31	
August 1 - 31	August 1 - 31	August

GOAL AND OBJECTIVES

The goal of this study was to conduct a creel survey of the North Coast (Areas 3 and 4) recreational fishery following the design and methodology used recently by DFO (Van Tongeren and Winther 2010), for the purpose of producing estimates of effort and catch of all species targeted by the recreational fishery. The creel survey included aerial effort counts and access point angler interviews, conducted throughout Areas 3 and 4 from June through August 2018. The precision goal for the study was to estimate total harvests of large components (such as Chinook Salmon) within 15% of the true value 19 times out of 20. Creel survey strata included temporal separation by month (or period) and day type (weekdays vs. weekend days). Spatial stratifications were based on creel survey subareas. All local fishing areas (i.e., those accessed from major boat launches in the area) were included in our survey design.

The seven objectives for this study were:

1. Maintain software for data management and analysis comparable to DFO's CREST program;
2. Provide estimates of the catch and effort of all vertebrate species targeted by the recreational fishery and associated estimates of precision;
3. Provide DFO with monthly in-season catch estimates for all salmon species and Halibut harvested in the North Coast recreational fishery;
4. Provide communication materials to recreational harvesters to convey the usefulness and importance of their catch data in fisheries management;
5. Collect biological data, such as scale samples and fin clip incidence for Coho and Chinook salmon, and Pacific Halibut lengths from a representative subsample of the landed harvest (during angler interviews);
6. Produce a final written report to be available to a broad audience; and
7. Present creel survey findings at the North Coast salmon post-season review meeting to a broad audience of commercial, recreational interests and First Nations.

Objective 1 was addressed over the course of the study period, and included the unforeseen need to stratify data by management period instead of by month, which required recoding of a considerable proportion of the analytical script. Objectives 2 and 6 were met by completing this report. Objective 3 was met throughout the study period by providing timely and complete summaries of the monthly survey activities (with complete creel analyses done for each update report). Objective 4 will be met subsequent to the publication of this report, as its contents will be based on the results presented herein. Objective 5 was met by biosampling fish at access points throughout the study period: fish were measured and sexed, scale samples and fin clip

incidence data were collected for Coho and Chinook salmon, heads were collected from adipose-clipped fish, and all biological data were submitted to DFO at the end of study period. Objective 7 was met in December 2018.

METHODS

A complete census of catch and effort was available for lodge-based anglers. However, the remainder of the recreational fishery occurred over too large an area to completely census catch and effort. Instead, statistically robust methods were used to estimate catch by multiplying estimates of *angler effort* by estimates of *catch per unit of effort*. Separate estimates were generated for each creel monitoring subarea, month, day type (weekday vs. weekend), and species.

For each creel subarea during each period and day type, fishing effort was estimated by counting boats that were actively fishing during overflight surveys (sub-area counts were reduced by the number of lodge-based boats thought to have been visible to the aircraft). Catch per unit of effort for non-lodge boats was estimated from interviews (see data forms in Appendix 1) conducted at known access points. During interviews, anglers were asked about their catch, effort, and fishing locations. They were also asked about their hourly fishing activity patterns.

Data collected during interviews included:

- Date;
- Number of anglers in party;
- Whether or not the trip was guided by a professional;
- Subarea angler activity – for each subarea, the hours during which angling activity was conducted;
- Subarea catch – the number of fish kept and the number released by species and subarea;
- Subarea gear – the amount and type of fishing gears used to catch each species in each subarea;
- Subarea targets – the number of hours spent targeting each species in each subarea (anglers that targeted only shellfish were noted);
- Biosampling – if allowed by the anglers, harvested fish were measured and sexed. Chinook and Coho salmon were checked for adipose clips and, if clipped and permission given, heads were collected for the Salmon Head Recovery Program. Scale samples were taken from Chinook and Coho salmon. Chinook Salmon were also examined for flesh colour;
- Refusals – whether or not the anglers refused an interview, allowed catch to be observed, or allowed catch to be biosampled; and
- Weather conditions.

The analytical methods used, similar to those used in Robichaud et al. (2017) and Robichaud and Addison (2018), provided a statistically unbiased estimate of catch per effort, assuming the anglers interviewed were representative of the entire non-lodge recreational fishery. Interviews were conducted at the three main access locations: Port Edward and Rushbrooke boat launches, and Prince Rupert Yacht Club. The locations surveyed were selected from all available access points, based on their geographical distribution and the amount of fishing activity that was

conducted from that site in recent years (see Van Tongeren and Winther 2010). Wherever possible, the busiest access points were selected preferentially in order to obtain the maximum number of interviews. This approach was based on two important observations: 1) the variability in CPE (catch-per-effort) among fishing parties landing at a single access point tends to be as great as the variability in CPE among different access points within a geographic area; and 2) CPE and effort can vary substantially both within and between days at a single site (English et al. 2002). Under these conditions it is better to obtain a large number of interviews covering all temporal strata for a small number of sites than to sample a larger number of sites and obtain fewer interviews and less complete temporal coverage for any specific site. The frequency of interview shifts was planned with the target of sampling 20% of estimated effort, while attempting to have “hands on” and direct observations of as many fish as possible meant allocating adequate time to each interview. The interview schedule was designed to capture data from representative angler parties in both day types, and over all time periods of the day. Ideally, interviews would capture data about angler activity in all creel subareas.

Interviewing sessions were to be separated into AM and PM shifts, with AM shifts occurring between 7:00 and 15:00 and PM shifts from 13:00 to 21:00. In total, 176 shifts were worked.

Fishing effort was estimated from boat-counts made during overflights. Overflights were conducted on 2-3 weekend days and usually on 5-6 weekdays each month (mid-season, the flights were partitioned into four periods rather than three months, which meant that period-specific sample sizes were lower than what we expected we would have when we were planning to generate *monthly* effort estimates). Timing of the overflights was targeted for peak fishing periods, and, although analytical methods allow for the occasional off-peak flight (e.g., due to weather or scheduling conflicts), all flights in 2018 coincided with peak fishing activity.

Angler Activity Patterns

Two weighting factors were used together with the interview-derived angling activity data to estimate the daily non-lodge fishing activity pattern (English et al. 2002). The first weighting factor, $W1$, expanded the numbers of days spent interviewing at each landing site to account for the total number of days available for sampling. That is, it was assumed that the daily activity pattern recorded during the interview shifts at landing site a , were consistent for landing site a , even during the days when no interviews occurred. A specific $W1$ was calculated for each landing site during each period and day type:

$$W1_{pda} = \frac{N_{pd}}{K_{pda}} \quad (\text{Equation 1})$$

where N_{pd} was the total number of type d days in period p ; and K_{pda} was the number of days during which interviews occurred at landing site a , on type d days during period p .

The second weighting factor, $W2$, expanded the numbers of interviews conducted, to account for the boats that were *not* interviewed (or that refused interviews). That is, it was assumed that the activity pattern recorded during the interview shifts also held for those anglers that were not interviewed. A specific $W2$ was calculated for each surveyor (u) operating on each surveying date (k) at each landing site (a):

$$W2_{aku} = \frac{LF_{aku} + LM_{aku} \left(\frac{LF_{aku}}{LF_{aku} + LN_{aku}} \right)}{LF_{aku} - ri_{aku}}, \quad (\text{Equation 2})$$

where LF_{aku} was the number of interviewed boats that were fishing (a subset of these, ri_{aku} , refused an interview, or had an incomplete interview that could not be used), LN_{aku} was the number of interviewed boats that were not fishing, and LM_{aku} was the number of boats that were not interviewed by surveyor u , during surveying date k , at landing site a . This equation incorporates the assumption that interviewed boats were representative, allowing us to estimate the portion of missed boats that were fishing.

We used the term $A_{pdsakut}$ to denote the number of non-lodge boats reporting activity during time-block t , as interviewed by surveyor u on survey date k , at landing site a , in subarea s , during period p , and on day type d (n_{pdsaku} was used to denote the total number of boats that reported activity during *any* time-block). The two correction factors were applied, and the data were summed over surveyors, survey dates and landing sites:

$$A'_{pdst} = \sum_k \sum_u \sum_a (W1_{pda} \cdot W2_{aku} \cdot A_{pdsakut}). \quad (\text{Equation 3})$$

Summing the adjusted number of anglers over the hourly time-blocks gave:

$$T'_{pds} = \sum_t A'_{pdst}. \quad (\text{Equation 4})$$

The proportion of non-lodge boats (P_{pdst}) that were actively fishing during each of the hourly time-blocks (t) was calculated for each period, day type, and subarea:

$$P_{pdst} = \frac{A'_{pdst}}{\sum_k \sum_u \sum_a (W1_{pda} \cdot W2_{aku} \cdot n_{pdsaku})}. \quad (\text{Equation 5})$$

Using this method, 96 unique angler activity patterns were to be estimated (i.e., 4 periods $\times$ 2 day types $\times$ 12 subareas). To reliably describe angler activity, a relatively large number of boats (~60) needed to be interviewed in each of the 96 blocks. In the end, some blocks contained too few interviews (Table 2), so it was decided to pool activity data over subarea. The equation for non-lodge angler activity was thus:

$$P_{pdt} = \frac{\sum_s A'_{pdst}}{\sum_s \sum_k \sum_u \sum_a (W1_{pda} \cdot W2_{aku} \cdot n_{pdsaku})}, \quad (\text{Equation 6})$$

and the associated variance was:

$$S^2_{P_{pdt}} = \frac{(P_{pdt})(1 - P_{pdt})}{\sum_s \sum_k \sum_u \sum_a (W1_{pda} \cdot W2_{aku} \cdot n_{pdsaku})}. \quad (\text{Equation 7})$$

Table 2. The amount of data (number of complete interviews) available to estimate non-lodge angler activity patterns and CPE, for all levels of each factor, 2018.

		Subarea											
Period	Day Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
June 1 - 15	WD	2	5	5	34	8	22	12	1	21	4	13	0
	WE	0	3	1	27	11	26	9	0	23	3	13	5
June 16 - July 9	WD	16	12	7	70	29	36	11	8	48	9	36	16
	WE	5	11	6	82	12	28	3	2	30	9	38	7
July 10 - 31	WD	8	9	5	43	36	53	14	5	92	6	23	20
	WE	4	11	5	30	28	31	15	5	60	7	22	6
August 1 - 31	WD	18	25	14	49	30	43	37	1	72	0	19	20
	WE	15	22	25	62	15	40	14	0	38	4	17	11

Catch Per Effort Estimation

Catch per effort (numbers of fish kept or released per boat trip) of non-lodge boats was estimated for each species from interviews of non-lodge anglers. Catch per effort was calculated separately for guided and non-guided boats, and overall; but only the overall estimates were used in subsequent calculations. For each interview (i), the guided status (g), period (p), and the catch (C) of each species (r) in each subarea (s) was recorded. Given that each interview represented one boat trip, then catch per effort (CPE), in terms of catch per boat trip, was calculated as:

$$CPE_{psrgi} = \frac{C_{psrgi}}{1} \quad \text{where } g \in \{\text{all, guided, not-guided}\}. \quad (\text{Equation 8})$$

Mean CPE was calculated as the average catch, over all n_{psg} boat trips:

$$\hat{CPE}_{psrg} = \frac{\sum_{i=1}^{n_{psg}} C_{psrgi}}{n_{psg}}. \quad (\text{Equation 9})$$

The variance for the estimate of mean catch per effort was calculated as:

$$S^2_{\hat{CPE}_{psrg}} = \frac{\sum (CPE_{psrgi} - \hat{CPE}_{psrg})^2}{(n_{psg} - 1)}. \quad (\text{Equation 10})$$

In the event that a period/subarea-specific sample size was < 5 , adjacent subareas would be pooled together as follows:

- if n for 3I, 3J or 3K was too low, then it was combined with the other two subareas, along with data from ambiguous parts of Area 3 ("3?") to give a 3?IJK regional average CPE;
- if n for 4A, 4B or 4C was too low, then it was combined with the other two to give a 4ABC regional average CPE;
- if n for 4D, 4E or 4F was too low, then it was combined with the other two to give a 4DEF regional average CPE; and
- if n for 4G, 4H or 4L was too low, then it was combined with the other two to give a 4GHL regional average CPE.

Similar methods were used to calculate the average number of fish harvested (retained) per boat trip (HPE), and the average number of fish released per boat trip (RPE).

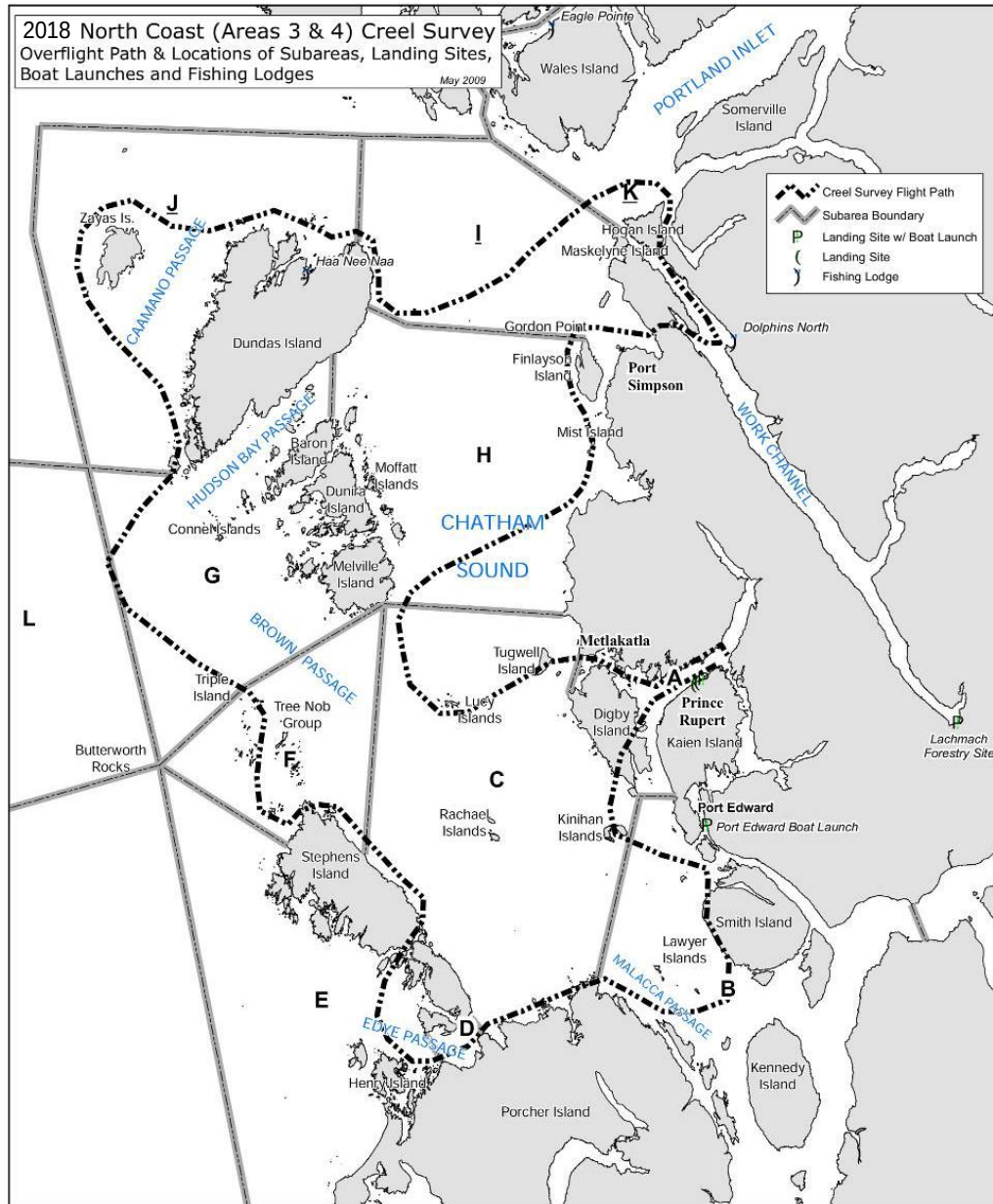


Figure 2. Map showing planned flight path of effort survey overflights. Letters indicate subareas. Flights originated and ended in Prince Rupert, and flew in the clockwise direction. On good weather days, the flights also included a foray into subarea 4L (path not shown).

Table 3. Summary of overflight sampling effort (number of flights shifts) by day type and by month (left) or period (right), 2018.

Month	Weekday	Weekend/ Holiday		Total	Period	Weekday	Weekend/ Holiday		Total
		Weekend/	Holiday				Weekend/	Holiday	
June	4	4	4	8	June 1 - 15	2	2	2	4
July	4	4	2	6	June 16 - July 9	3	2	2	5
August	6	6	3	9	July 10 - 31	3	2	2	5
Total	14	14	9	23	August 1 - 31	6	3	3	9
					Total	14	9	9	23

Angler Effort Estimation

To obtain statistically valid estimates of non-lodge fishing effort, boats were counted during overflight surveys conducted from a fixed-wing aircraft traveling along a set path over the study area (see Figure 2). In order to calculate the non-lodge effort, the estimated counts of active lodge boats needed to be subtracted from the overflight counts for that particular day.

Lodge Effort Census

A complete census of lodge-based angling effort was collected from lodge log books by DFO personnel (I. Winther, DFO Prince Rupert, pers. comm.). DFO used the following procedure to estimate the number of active lodge vessels that would have been available to be observed during the flights: 1) Lodge based vessels were considered unavailable to the flights on change-over days (days when the lodges received new clients; on typical change-over days, departing clients leave before the flight and new clients arrive after the flight). Change-over days occurred twice a week but timing was often different for each lodge, making 29% (2/7) of the lodge effort not visible to flights due to change-over days. 2) Lodges reported their lunchtime activity to DFO, which ranged from virtually always going in for a mid-day break to rarely leaving the water during the day. Thus, approximately 60% of the remaining vessel activity was available to the flight counts on ‘regular’ days (i.e., not change-over). 3) Vessels on mid-day break or on change-over days were excluded from the numbers of vessels visible to the flights. All of the lodge catch was reported from Area 3, without subarea-specific effort information. Thus, lodge effort was partitioned among the Area 3 subareas, see below.

Non-lodge Effort Surveys

Table 3 shows the number of overflight surveys scheduled for each day type and period. Each survey was designed to allow visual inspection of the subareas through which the aircraft passed. Each time the aircraft entered a subarea, the start and end times were recorded, along with the numbers of small (< 30 ft) and large (≥ 30 ft) recreational vessels that were observed, tallied separately depending on whether or not they were actively fishing at the time.

During survey o (conducted during period p and on day type d), observers tallied the total number of recreational boats that were actively fishing at time t on their i^{th} pass over subarea s , V_{pdsoit} . For each visit to a subarea within Management Area 3, the relative proportion of boats was calculated, ρ_{pdsoit} , and the total number of lodge boats (λ_{pdo}) apportioned accordingly:

$$\lambda_{pdsoit} = \rho_{pdsoit} \cdot \lambda_{pdo} \quad . \quad (\text{Equation 11})$$

The number of non-lodge boats was hence:

$$v_{pdsoit} = V_{pdsoit} - \lambda_{pdsoit} \quad . \quad (\text{Equation 12})$$

Since angling occurs over the course of the entire day, the number of non-lodge boats that were observed was divided by the proportion of average daily number of boats active (P_{pdst}) during the time block when the observations were recorded. Since the flight path (Figure 2) brought the observers into some subareas more than once during the survey (covering different ground, thus the tallies were considered additive), the adjusted tallies were summed over all visits, to calculate the total number of active non-lodge fishing boats on the day of the survey, by subarea, B_{pdso} :

$$B_{pdso} = \sum_i \frac{v_{pdsoit}}{P_{pdst}} . \quad (\text{Equation 13})$$

The variance of B_{pdso} was calculated using the standard formulas for combining the variance of products and quotients of two independent random variables (Goodman 1960):

$$\begin{aligned} \text{if } z = x / y, \text{ } Var(z) &= (y^{-2})Var(x) + (x^2 y^{-4})Var(y) ; \\ \text{if } z = xy, \text{ } Var(z) &= (y^2)Var(x) + (x^2)Var(y) ; \text{ and} \\ \text{if } z = cx, \text{ } Var(z) &= (c^2)Var(x) \text{ where } c \text{ is a constant.} \end{aligned} \quad (\text{Equation 14})$$

Since there was no variance associated with the boat counts,

$$S_{B_{pdso}}^2 = (v_{pdsoit})^2 (P_{pdst})^{-4} S_{P_{pdst}}^2 . \quad (\text{Equation 15})$$

Next, the survey-specific ‘daily effort’ estimates were averaged over the number of surveys conducted, n_{pds} , as:

$$\hat{B}_{pds} = \frac{\sum_{o=1}^{n_{pds}} B_{pdso}}{n_{pds}} , \quad (\text{Equation 16})$$

with a variance of:

$$S_{\hat{B}_{pds}}^2 = (n_{pds})^{-2} S_{B_{pdso}}^2 . \quad (\text{Equation 17})$$

Total fishing effort for each sub-period was calculated for each day type and subarea by multiplying the period-specific average daily effort (where $\hat{B}_{pds}$ for Period 2 was applied to Sub-periods 2 and 3, see Table 1) by the number days of day type d that occurred in sub-period c :

$$E_{cds} = \hat{B}_{pds} \cdot N_{cd} , \quad (\text{Equation 18})$$

which was then summed over day type to give:

$$E_{cs} = \sum_d \hat{B}_{pds} \cdot N_{cd} , \quad (\text{Equation 19})$$

with a variance of:

$$S_{E_{cs}}^2 = \sum_{d=1}^2 S_{\hat{B}_{pds}}^2 \cdot N_{cd}^2 . \quad (\text{Equation 20})$$

Total monthly fishing effort was calculated by summing over the appropriate sub-periods (total effort for June was the sum of effort in Sub-periods 1 and 2; that for July was the sum of Sub-periods 3 and 4; and August effort was equal to that in Sub-period 5). As always, variances for sums were additive (Goodman 1960).

Because of inclement weather, not all overflights surveyed every subarea. Especially for the more exposed subareas, it was possible to have no overflight data in a given combination of period and day type. In this event, subarea-specific effort was scaled using: 1) the proportion of effort in that subarea, relative to that in a three-subarea cluster (see pooling of adjacent subareas for CPE estimation, above), measured during overflights for which all three subareas were surveyed; and 2) the effort in the cluster’s other two subareas, estimated for the combination of period and day type during which the data gap occurred.

Catch and Harvest Estimation

Total catch was calculated for each sub-period, subarea and species by multiplying sub-period-specific angling effort by period-specific catch per effort, and then summing over day type:

$$C_{csr} = \sum_d (E_{cds} \cdot \hat{CPE}_{psrg}) \quad (\text{Equation 21})$$

where g = 'all' (see Equation 8). The standard errors for these catch estimates were:

$$S_{C_{csr}} = \sqrt{\sum_d \left(E_{cds}^2 \frac{S_{CPE_{psrg}}^2}{n_{psg}} + CPE_{psrg}^2 \frac{S_{E_{cds}}^2}{n_{pds}} + \frac{S_{CPE_{psrg}}^2}{n_{psg}} \frac{S_{E_{cds}}^2}{n_{pds}} \right)} \quad (\text{Equation 22})$$

where n_{pds} and n_{psg} were the sample sizes for E and CPE estimates, respectively. Total monthly catch was calculated by summing over the appropriate sub-periods (total catch for June was the sum of catch in Sub-periods 1 and 2; that for July was the sum of Sub-periods 3 and 4; and August catch was equal to that in Sub-period 5), with additive variances.

Similar methods were used to estimate numbers of fish harvested, and numbers of fish released, replacing CPE with HPE or with RPE in Equations 21 and 22.

Biosampling

When given permission by the anglers, observers collected biological data from Chinook and Coho salmon and Pacific Halibut. All three species were sexed when they were intact. Chinook and Coho salmon were examined for clipped adipose fins (mark rate data used by DFO for coded wire tag analyses) and potentially head tags. The flesh color was noted for Chinook Salmon. Scale samples were collected from Chinook and Coho salmon (used by DFO to determine age, and stock composition).

Ideally, all anglers would allow their catch to be biosampled, and that the examined fish were an unbiased sample of all harvested fish. But Robichaud et al. (2016) theorized that the interview protocol, used until the end of 2015, may have resulted in bias. The protocol had observers ask anglers about adipose clips before asking permission to observe the fish. Anglers were often engaged in a conversation about the purpose of the fin-clips and how the data would be used, and it appeared that anglers with clipped fish were more likely to permit handling and biosampling of catch than when no clipped fish were present. To avoid this bias, we modified our data collection protocol in 2016 so that anglers were asked if their catch could be biosampled before any other interactions took place. Also starting 2016, no fin clip data were accepted on the word of the anglers: the only fin-clip data we used were those which came from observers when they handled and inspected the catch.

Biomass of Harvested Pacific Halibut

Halibut lengths were converted to weights using the International Pacific Halibut Commission (Clark 1992) regression equations:

$$\begin{aligned} \text{Weight(lb)}_{\text{net}} &= 6.921 \times 10^{-6} \times \text{FL(cm)}^{3.24} \\ \text{Weight(lb)}_{\text{round}} &= 9.166 \times 10^{-6} \times \text{FL(cm)}^{3.24} \end{aligned} \quad (\text{Equation 23})$$

Converting average length to average weight allowed harvested biomass to be calculated from harvested fish counts.

Statistical Analyses

For each fish taxa, differences in CPE among periods and between trip-types (guided vs. non-guided) were examined using Kruskal Wallis tests (Zar 1984). The same tests were used to examine the effects of period and day type on effort, and the effects of month on catch and harvest. When statistically significant effects were observed, *post-hoc* Kruskal-Wallis tests were used to compare between each possible pair, where the false discovery rate was controlled using the Benjamini–Hochberg procedure (Benjamini and Hochberg 1995).

Trailer Census

Starting at the beginning of May, the total number of parked trailers at the landing locations was recorded daily during the peak of the activity timing window (i.e., between noon and 2 PM; similar to the targeted timing of aerial overflights). Theoretically, trailer counts could be used as an index of fishing effort, are more economical than overflights, and are not weather dependant, though no sub-area-specific effort data are collected. DFO has been collecting trailer count data for years and we are maintaining the time series.

Linear regression was used to determine the relationship between trailer counts and fishing effort. A data point was generated each time a trailer census occurred on the same day as an overflight. The location-specific trailer counts and non-lodge boat counts were used as the independent and dependent variables, respectively.

Trailer count data were collected throughout May in order to calculate an approximate index of effort, to test whether substantial angling activity is being missed by starting the creel survey in June each year. We used the above linear regression to calculate what the daily boat counts *would have been* had there been overflights each day. Then, the activity pattern data were used to expand the estimated boat counts to calculate total fishery effort by day (trailer count data were grouped into clusters of days to determine with activity profile best suited the data). Summing over months permitted comparisons of estimated effort in May 2018 to that during the three survey months.

RESULTS

Angler Interviews

Over the three month study period, there were 176 observer shifts, during which 1,482 interviews were conducted, involving 4,410 non-lodge anglers (Table 4; Figure 3). July was the busiest month, with 605 interviews conducted (average 10.1 interviews per shift), which represented 41% of the total number of study period interviews (Table 4).

Non-lodge boat trip activity was unevenly distributed among the subareas, with 4F being the overwhelming favourite (Table 5). Subareas 4A, and 4C were the next most likely areas for anglers to have reported activity. The subareas with the lowest number of trips reporting activity were 4E and 4G. Only 61 interviews were of parties that only targeted shellfish, and those were largely in subarea 4A (Table 6).

Table 4. Monthly number of shifts, and numbers of boats interviewed per landing site, 2018. Non-lodge effort is expressed in units of boat trips. Sampling efficiency is also calculated.

Landing Site	June				July				August				TOTAL			
	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint
Port Edward Boat Launch	90	226	20	4.5	207	600	20	10.4	119	340	19	6.3	416	1166	59	7.1
Prince Rupert Yacht Club	99	333	19	5.2	165	538	20	8.3	122	405	19	6.4	386	1276	58	6.7
Rushbrooke Boat Launch	209	597	20	10.5	233	681	20	11.7	238	690	19	12.5	680	1968	59	11.5
TOTAL	398	1,156	59	6.7	605	1,819	60	10.1	479	1,435	57	8.4	1,482	4,410	176	8.4
% of Total Interviews		27%				41%				32%						
Estimated Non-lodge Effort		2,296				4,017				3,054				9,367		
% of Non-lodge Effort		25%				43%				33%						
% of Effort Sampled		17%				15%				16%				16%		

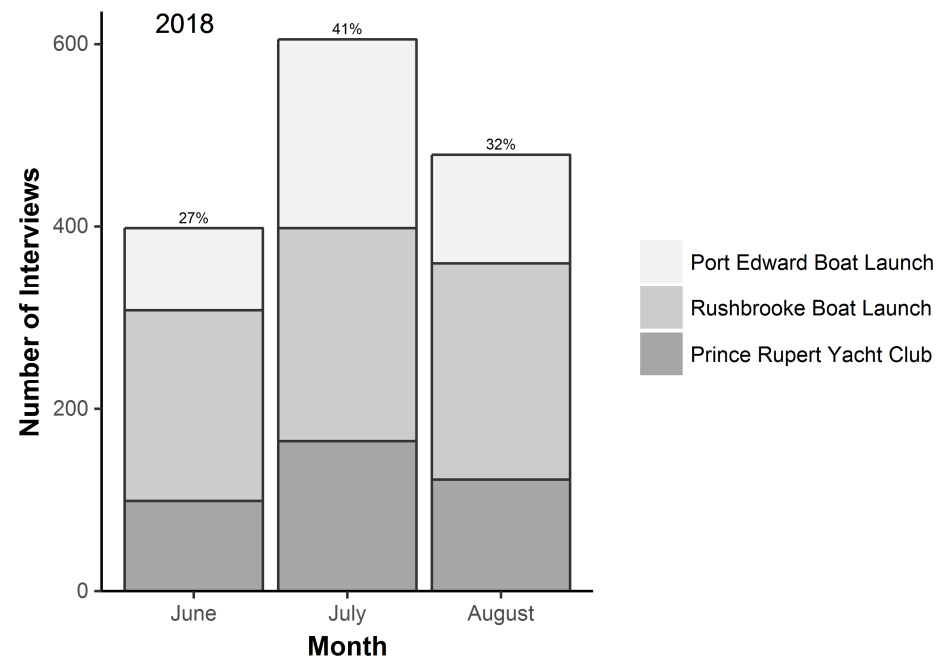


Figure 3. Monthly number of interviews conducted at each of three landing sites, 2018.

Table 5. Fishing locations as reported by interviewed non-lodge parties, by month and subarea, 2018. For interviews that reported activity in multiple subareas, the boat trip tally was evenly partitioned among the subareas visited.

Month	Subarea															TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	3?	4?	Refusal	
June	7	10	10	87	32	60	23	7	48	13	51	13	1	11	26	398
July	20	25	14	73	67	91	27	9	150	14	54	26		12	24	605
August	27	34	31	67	37	67	43	1	91	2	28	25		5	22	479
TOTAL	54	69	55	227	136	217	93	17	289	30	133	64	1	28	72	1,482

Table 6. Numbers of non-lodge boat trips that reported only fishing for shellfish, by month and subarea, 2018.

Month	Subarea															TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	3?	4?	Refusal	
June	0	0	0	24	3	0	0	0	1	0	1	0	0	0	-	28
July	0	0	0	9	1	0	0	0	0	0	0	0	0	0	-	10
August	1	0	0	18	1	0	0	0	0	2	1	0	0	0	-	23
TOTAL	1	0	0	51	5	0	0	0	1	2	2	0	0	0	-	61

Table 7. Numbers of guided (i.e., led by professional guides) and unguided boat trips, by landing site, 2018.

Landing Site	Trip Type		% guided
	Guided	Unguided	
Port Edward Boat Launch	5	411	1.2%
Prince Rupert Yacht Club	235	151	61%
Rushbrooke Boat Launch	253	427	37%
TOTAL	493	989	33%

On average, 33% of the non-lodge boat trips were guided by a professional. The guiding rate varied by landing site. Only 5 of 416 trips (1.2%) interviewed at Port Edward were guided, whereas trips departing from Rushbrooke and Prince Rupert Yacht Club were guided 37% and 61% of the time, respectively (Table 7).

In all there were 72 refusals (Table 5), or 4.9% refusal rate. The highest refusal rate (8.8%) was at Rushbrooke Boat Launch, where 83% of the refusals occurred. The refusal rates for Prince Rupert Yacht Club and Port Edward Boat Launch were 1.0% and 1.9%, respectively.

Angler Activity Patterns

Non-lodge angler activity patterns estimates varied little among periods (Figure 4). There was virtually no difference between the weekday and weekend/holiday activity patterns. During the July 10-31 period, weekend anglers were more likely to be activity during the peak times, as compared to weekday anglers; whereas the opposite pattern held for the other three periods. Timing of our overflights lined up with peak activity levels in all periods.

Catch Per Effort Estimates

Sample sizes were adequate ($n \geq 5$) to produce reliable estimates of non-lodge CPE in most combinations of periods and management areas, (Table 8) except in subareas 3I and 4E from June 1-15, and in subareas 4E and 4G in August. However, when the interview data were split between guided and non-guided groups, period-specific sample sizes did not suffice in several of the management areas (i.e., 3I-3K, 4B, 4D-E, 4G, and 4L; Table 8). In those cases where sample sizes were limited, data from three adjacent subareas were pooled together for CPE calculations (Table 9). Average number of fish harvested per non-lodge boat trip (HPE) was calculated for each species by period, subarea, and by trip type (Appendix 2).

HPE varied significantly among periods (Table 10) for Chinook ($\chi^2 = 26.0$, $df = 3$, $P < 0.0001$), Chum (*O. keta*, $\chi^2 = 21.9$, $df = 3$, $P < 0.0001$), Coho ($\chi^2 = 28.3$, $df = 3$, $P < 0.0001$), and Pink salmon (*O. gorbuscha*, $\chi^2 = 18.1$, $df = 3$, $P = 0.0004$). For Chinook Salmon, median HPE from June 16-July 9 was significantly lower than in any of the other three periods (Table 10; Figure 5). For Chum Salmon, median HPE from August was significantly higher than in any of the other three periods (Table 10; Figure 5). For Coho Salmon, median HPE from June 1-15 was significantly lower than all other periods, median HPE from June 15-July 9 was significantly lower than the following two periods, and the last two periods did not differ significantly (Table 10; Figure 5). For Pink Salmon, HPE from June 1-July 9 was significantly lower than that from July 10 onwards (Table 10; Figure 5). There were no significant differences among periods for HPE of Sockeye salmon (*O. nerka*), or for any species of groundfish (including Pacific Halibut and Lingcod; Figure 5). For Tiger Rockfish (*Sebastes nigrocinctus*), median HPE from July 10-31 was significantly lower than in August, and no other pairwise between-period difference were significant ($\chi^2 = 8.8$, $df = 3$, $P = 0.033$; Table 10; Figure 5). No other rockfish species showed significant among-period differences in the number of fish harvested per boat trip.

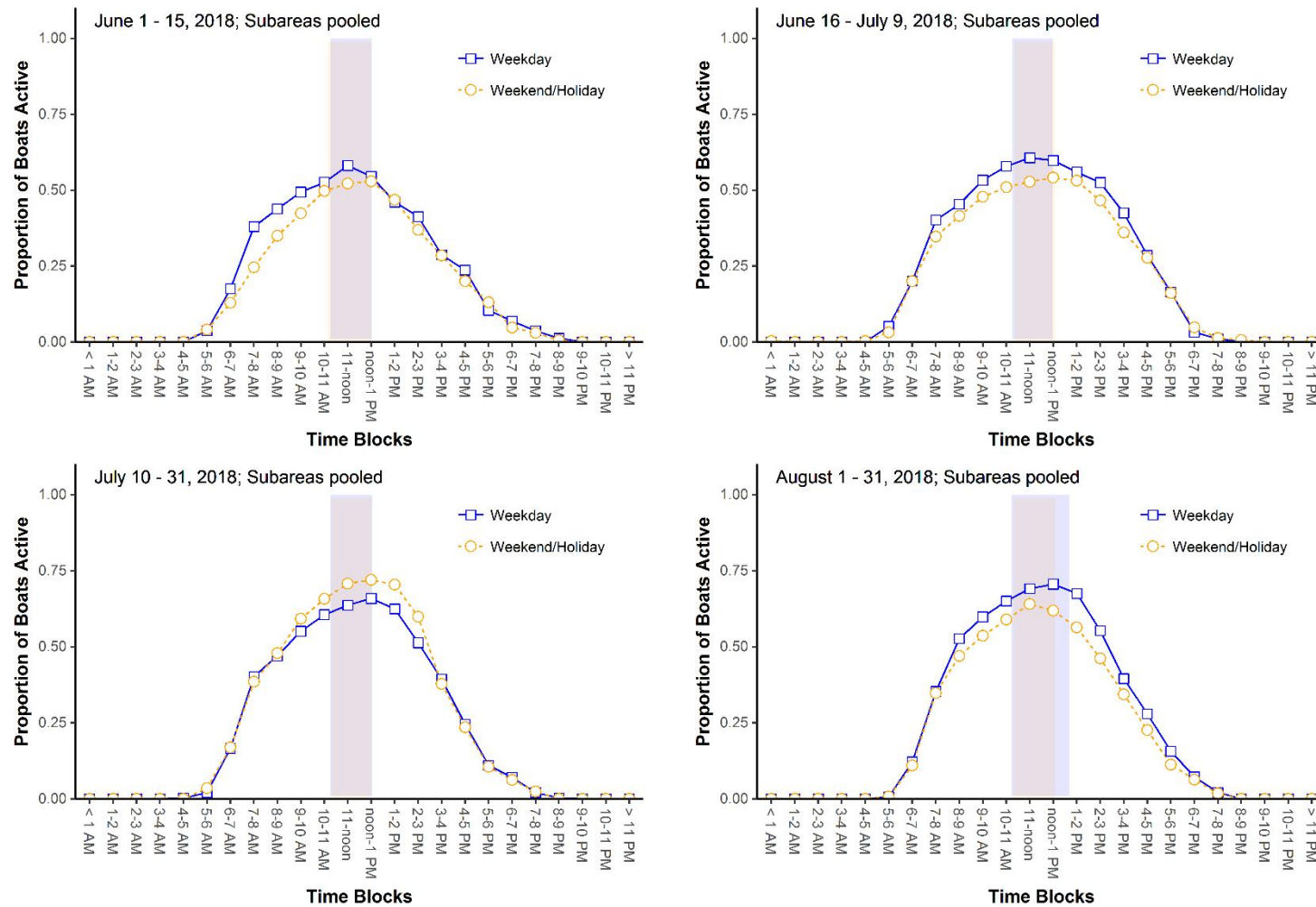


Figure 4. Angler activity pattern (proportion recreational angling boats that were actively fishing during each hour-long time block) by day type and period, pooled over all subareas, 2018. Shaded regions bound the earliest and latest flight times for each day type (blue = weekday; orange = weekend/holiday) and period.

Table 8. Sample sizes for CPE (harvest per effort) estimates by period, for guided and not guided non-lodge boat trips, 2018.

		Subarea											
Period	Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
June 1 - 15	Guided	0	6	5	35	1	13	4	0	12	4	11	3
	Unguided	2	2	1	26	18	35	17	1	32	3	15	2
	Combined	2	8	6	61	19	48	21	1	44	7	26	5
June 16 - July 9	Guided	9	10	3	89	0	13	4	3	44	6	40	21
	Unguided	12	13	10	63	41	51	10	7	34	12	34	2
	Combined	21	23	13	152	41	64	14	10	78	18	74	23
July 10 - 31	Guided	4	9	2	35	1	9	0	0	52	8	17	20
	Unguided	8	11	8	38	63	75	29	10	100	5	28	6
	Combined	12	20	10	73	64	84	29	10	152	13	45	26
August 1 - 31	Guided	13	26	13	42	1	12	4	0	46	1	12	26
	Unguided	20	21	26	69	44	70	47	1	64	3	24	5
	Combined	33	47	39	111	45	82	51	1	110	4	36	31

Table 9. Subarea data (pooled over several adjacent areas, in some cases) used to calculate CPE in each period subarea, for guided and not guided non-lodge boat trips, 2018.

Subarea													
Dates	Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
June 1 - 15	Guided	3?,I,J,K	3J	3K	4A	4A,B,C	4C	4D,E,F	4D,E,F	4F	4G,H,L	4H	4G,H,L
	Unguided	3?,I,J,K	3?,I,J,K	3?,I,J,K	4A	4B	4C	4D	4D,E,F	4F	4G,H,L	4H	4G,H,L
	Combined	3?,I,J,K	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4L
June 16 - July 9	Guided	3I	3J	3?,I,J,K	4A	4A,B,C	4C	4D,E,F	4D,E,F	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4G,H,L
	Combined	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
July 10 - 31	Guided	3?,I,J,K	3J	3?,I,J,K	4A	4A,B,C	4C	4D,E,F	4D,E,F	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
	Combined	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
August 1 - 31	Guided	3I	3J	3K	4A	4A,B,C	4C	4D,E,F	4D,E,F	4F	4G,H,L	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G,H,L	4H	4L
	Combined	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G,H,L	4H	4L

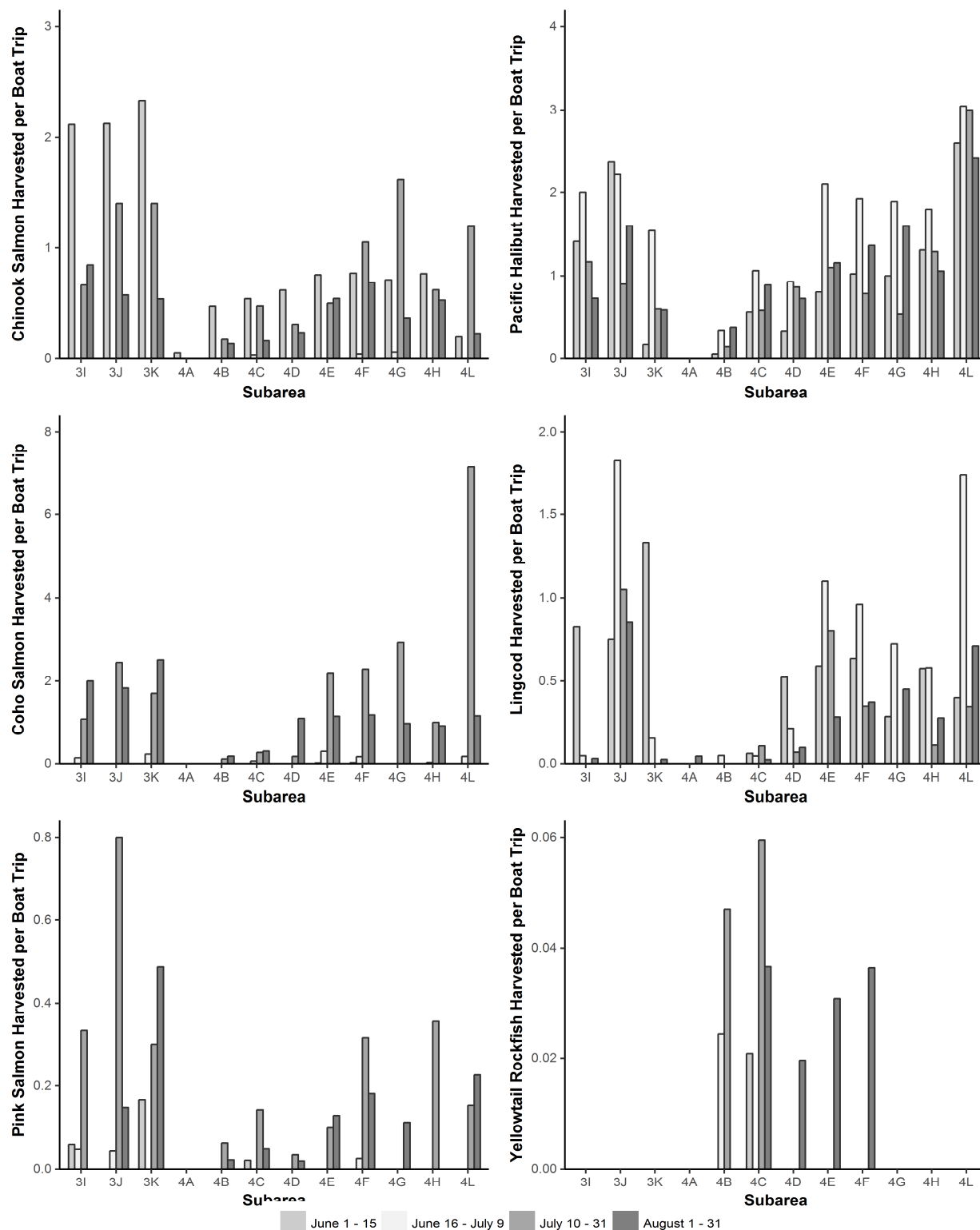


Figure 5. Average number of fish harvested per non-lodge boat trip, for six sportfish species, by period and subarea, 2018. Data are for guided and unguided trips combined. Note that Y axis scale varies among plots.

Table 10. Kruskal-Wallis tests for differences in HPE (fish harvested per non-lodge boat trip) among periods, by species, 2018. Subareas treated as replicates.

Species	Kruskal-Wallis Statistics			Median HPE			
	χ^2	df	P	Jun 1-15	Jun 16 - Jul 9	Jul 10-31	Aug 1-31
Chinook Salmon	26.0	3	<0.0001	0.7	0	0.6	0.4
Chum Salmon	21.9	3	<0.0001	0	0	0	0.03
Coho Salmon	28.3	3	<0.0001	0	0.04	1.4	1.1
Pink Salmon	18.1	3	0.0004	0	0	0.1	0.08
Sockeye Salmon	6.1	3	0.11	0	0	0	0
Arrow Flounder	2.0	3	0.56	0	0	0	0
Big Skate	6.1	3	0.11	0	0	0	0
Cabezon	6.7	3	0.08	0	0	0	0
English Sole	2.1	3	0.55	0	0	0	0
Greenling	1.2	3	0.75	0	0	0	0
Lingcod	5.2	3	0.16	0.6	0.4	0.09	0.2
Pacific Cod	3.6	3	0.31	0.08	0.06	0.01	0.02
Pacific Halibut	5.4	3	0.15	0.9	1.8	0.8	1.0
Ratfish	0.0	3	1	0	0	0	0
Rock Sole	2.0	3	0.56	0	0	0	0
Sablefish	3.2	3	0.35	0	0	0	0.010
Starry Flounder	0.0	3	1	0	0	0	0
Black Rockfish	5.7	3	0.13	0	0.03	0	0.1
Boccacio Rockfish	0.0	3	1	0	0	0	0
Canary Rockfish	6.1	3	0.11	0	0	0	0
China Rockfish	4.8	3	0.19	0	0	0	0
Copper Rockfish	3.3	3	0.35	0	0	0	0.007
Dusky Rockfish	2.0	3	0.56	0	0	0	0
Quillback Rockfish	5.2	3	0.16	0.02	0.1	0.04	0.2
Redband Rockfish	3.0	3	0.39	0	0	0	0
Redstripe Rockfish	3.0	3	0.39	0	0	0	0
Tiger Rockfish	8.8	3	0.033	0	0	0	0.005
Widow Rockfish	0.0	3	1	0	0	0	0
Yelloweye Rockfish	3.2	3	0.36	0	0	0	0
Yellowtail Rockfish	3.5	3	0.32	0	0	0	0

Guided trips had higher HPE than non-guided trips, and the differences were statistically significant for a variety of species (Table 11). In particular, guided trips caught 480% more Lingcod ($\chi^2 = 11.6$, $df = 1$, $P = 0.0007$) and 227% more Pacific Halibut ($\chi^2 = 17.7$, $df = 1$, $P < 0.0001$) than unguided trips. Although guided trips caught 222% more Chinook Salmon and 236% more Coho Salmon than unguided trips, the differences were not statistically significant (Chinook Salmon: $\chi^2 = 2.7$, $df = 1$, $P = 0.10$; Coho Salmon: $\chi^2 = 1.7$, $df = 1$, $P = 0.19$).

For lodge boat trips, HPE (and RPE for Chinook Salmon) estimates are shown in Table 12.

Table 11. Kruskal-Wallis tests for differences in HPE (fish caught per non-lodge boat trip) between guided and non-guided boats, by species, 2018. Periods and subareas treated as replicates.

Species	Kruskal-Wallis Statistics			Median HPE	
	χ^2	df	P	Guided	Not
Chinook Salmon	2.7	1	0.10	0.8	0.4
Chum Salmon	0.6	1	0.43	0	0
Coho Salmon	1.7	1	0.19	0.1	0.05
Pink Salmon	2.1	1	0.15	0	0
Sockeye Salmon	0.4	1	0.55	0	0
Arrow Flounder	4.1	1	0.042	0	0
Big Skate	2.0	1	0.16	0	0
Cabezon	3.0	1	0.09	0	0
English Sole	5.2	1	0.022	0	0
Greenling	0.0	1	0.83	0	0
Lingcod	11.6	1	0.0007	0.5	0.1
Pacific Cod	5.4	1	0.020	0.02	0
Pacific Halibut	17.7	1	<0.0001	1.7	0.7
Ratfish	0.0	1	1	0	0
Rock Sole	1.1	1	0.29	0	0
Sablefish	2.9	1	0.09	0	0
Starry Flounder	0.0	1	1	0	0
Black Rockfish	6.4	1	0.011	0.02	0
Boccacio Rockfish	0.0	1	1	0	0
Canary Rockfish	3.1	1	0.08	0	0
China Rockfish	2.1	1	0.15	0	0
Copper Rockfish	5.6	1	0.018	0	0
Dusky Rockfish	4.1	1	0.042	0	0
Quillback Rockfish	0.2	1	0.68	0.08	0.08
Redband Rockfish	1.0	1	0.32	0	0
Redstripe Rockfish	1.0	1	0.32	0	0
Tiger Rockfish	2.8	1	0.10	0	0
Widow Rockfish	0.0	1	1	0	0
Yelloweye Rockfish	5.7	1	0.017	0	0
Yellowtail Rockfish	3.0	1	0.08	0	0

Table 12. Average number of harvested fish (and for Chinook Salmon, released) per boat trip, from the census of Area 3 lodges, 2018.

Species	June	July	August	Total
Chinook Salmon:				
retained	0.5	0.6	0.6	0.6
released	1.0	0.7	0.4	0.8
Coho Salmon	0.02	1.4	2.6	1.1
Pink Salmon	0	0.02	0	0.009
Chum Salmon	0	0.02	0.05	0.02
Sockeye Salmon	0	0	0	0
Pacific Halibut	0.3	0.3	0.1	0.3
Lingcod	0.7	0.6	0.1	0.6
Other groundfish	0	0	0	0
Yelloweye Rockfish	0	0	0	0
Other rockfish	0.5	0.6	0.1	0.5

Angler Effort Estimates

Over the three month study period, a total of 23 effort surveys were conducted, including 14 weekday surveys and 9 conducted on weekends/holidays (Table 3, Appendix 3). The numbers of lodge boats that were visible during the overflights (Appendix 3) were partitioned among the subareas in Management Area 3, and subtracted from the overflight counts of small boats. The remaining small boat counts (“the non-lodge small boat counts”) were significantly correlated to trailer counts at both survey launch sites (Rushbrooke: $F_{1,20} = 46.2$, $P < 0.0001$; Port Edwards: $F_{1,20} = 61.4$, $P < 0.0001$), and when the two launch sites were combined ($F_{1,20} = 71.8$, $P < 0.0001$). The fit of the individual-sites regressions (Rushbrooke $R^2 = 0.68$; Port Edward $R^2 = 0.74$) were lower than that of the combined-site regression ($R^2 = 0.77$; Figure 6).

The non-lodge boat counts were adjusted for non-lodge angler effort patterns in order to estimate the total monthly non-lodge angling effort for each subarea in 2018 (Table 13, Figures 7 and 8). Median daily non-lodge effort on weekdays (5.3 trips per day per subarea) was significantly lower than on weekends/holidays (9.0; $\chi^2 = 4.3$, $df = 1$, $P = 0.038$). Effort also varied among periods, ranging from about 5.2 or 5.3 trips per day per subarea (in June 1-15 and in August),

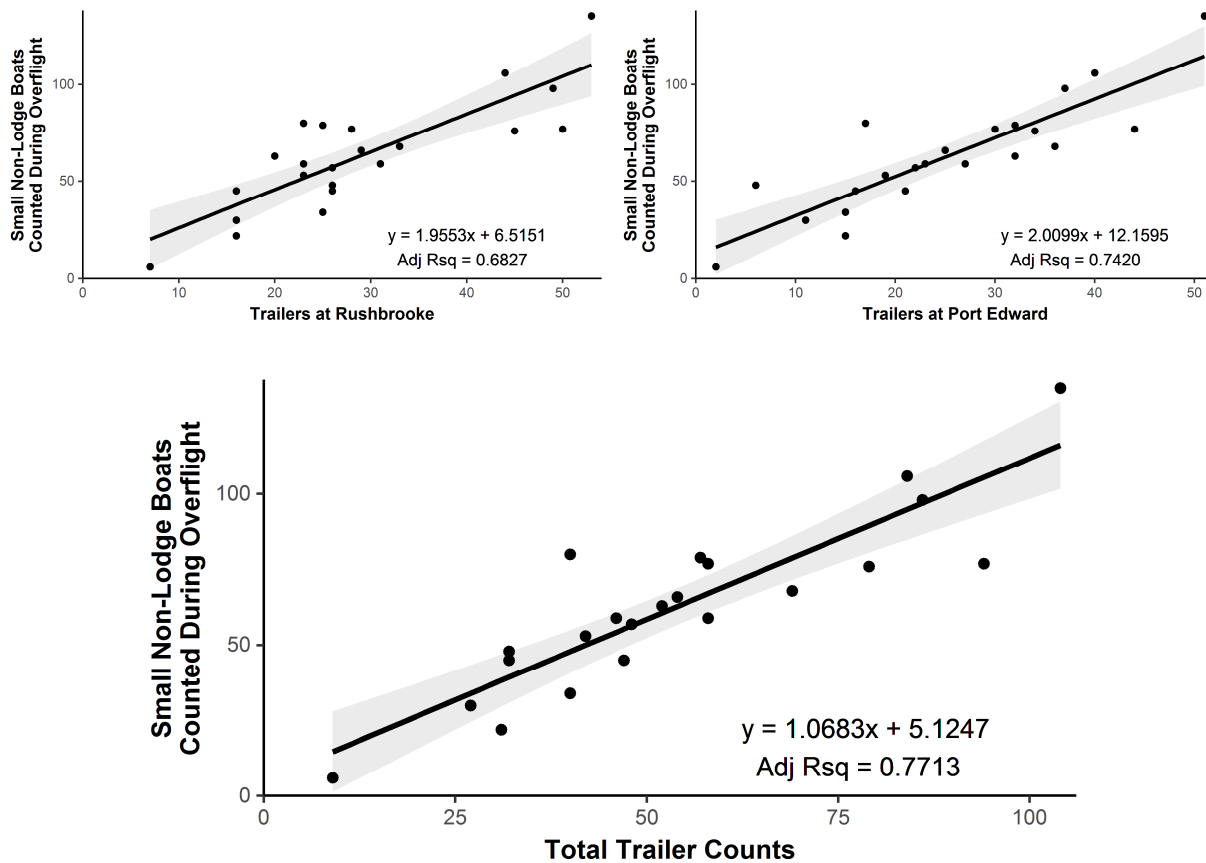


Figure 6. Linear regressions showing the relationships between small non-lodge boats counted during overflights and trailers concurrently counted at landing sites. Shaded area shows the 95% confidence interval of the predicted values. Note that X axis scale varies among plots.

Table 13. Lodge, non-lodge and total effort estimates (boat trips per month), by month and subarea, 2018.

Month		Subarea												TOTALS			SE of the Totals		
		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June	Creel Estimate	241.2	279.0	91.7	61.4	163.7	454.1	160.4	55.0	279.2	156.5	234.8	118.7	612	1,684	2,296	8	10	13
	Lodge Report	227.5	263.1	86.5										577	0	577			
	TOTAL JUNE	468.7	542.1	178.2	61.4	163.7	454.1	160.4	55.0	279.2	156.5	234.8	118.7	1,189	1,684	2,873	8	10	13
	SE	5.8	4.9	2.1	0.9	3.6	5.0	4.0	1.1	4.6	2.7	4.0	2.5						
July	Creel Estimate	415.9	404.6	252.7	142.4	171.7	902.2	63.5	108.9	933.0	168.5	299.6	154.5	1,073	2,944	4,017	8	15	17
	Lodge Report	223.2	217.1	135.6										576	0	576			
	TOTAL JULY	639.1	621.7	388.3	142.4	171.7	902.2	63.5	108.9	933.0	168.5	299.6	154.5	1,649	2,944	4,593	8	15	17
	SE	5.4	4.9	3.4	1.5	2.8	8.3	1.0	1.5	11.5	2.3	3.5	2.3						
August	Creel Estimate	377.1	363.5	472.9	121.3	131.0	576.7	56.4	72.7	497.0	127.0	126.7	132.0	1,213	1,841	3,054	8	10	13
	Lodge Report	77.7	74.9	97.4										250	0	250			
	TOTAL AUGUST	454.8	438.4	570.3	121.3	131.0	576.7	56.4	72.7	497.0	127.0	126.7	132.0	1,463	1,841	3,304	8	10	13
	SE	4.2	4.2	5.4	1.0	1.6	4.6	0.7	1.7	7.4	1.9	1.5	2.5						
FULL SEASON TOTAL		1,563	1,602	1,137	325	466	1,933	280	237	1,709	452	661	405	4,302	6,469	10,770	14	21	25
FULL SEASON SE		9	8	7	2	5	11	4	3	14	4	6	4						

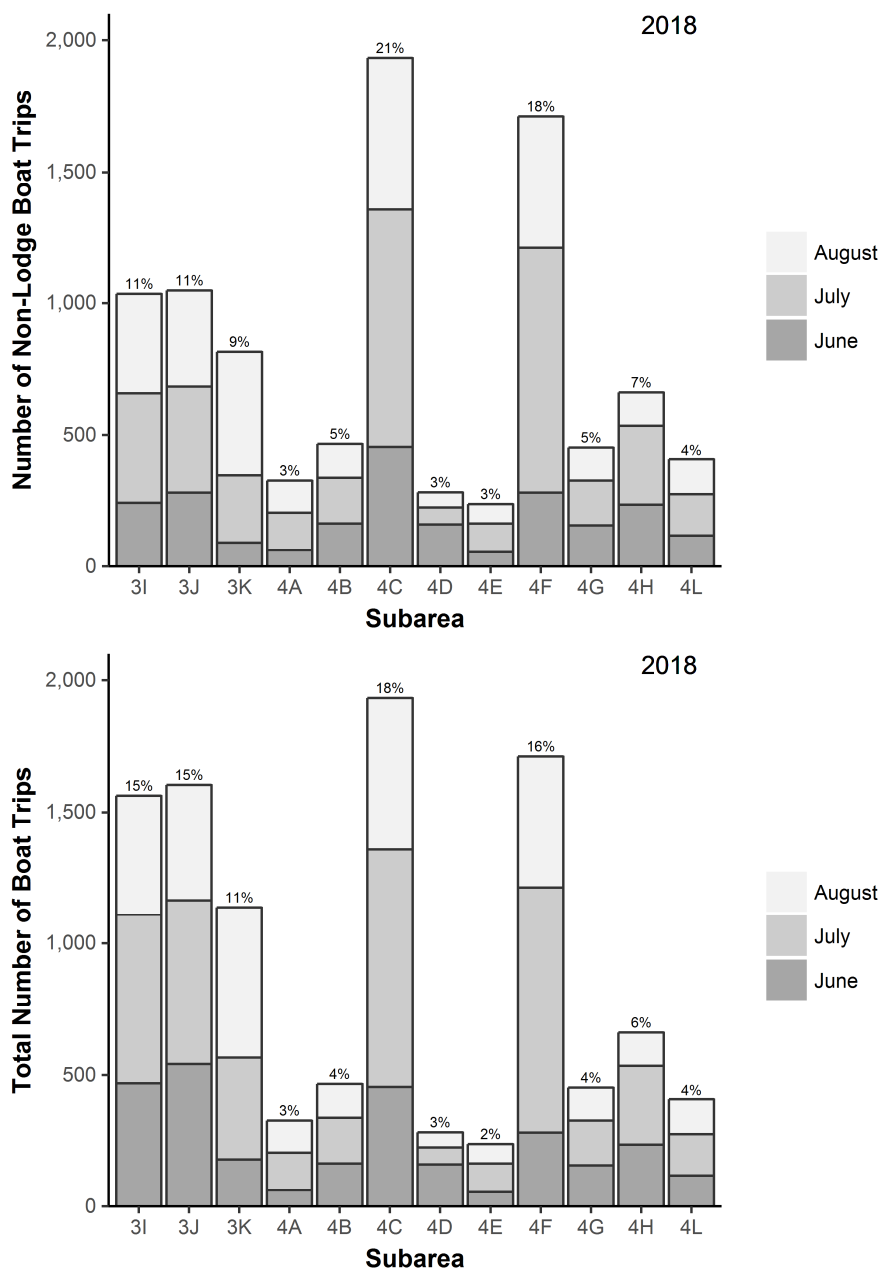


Figure 7. Total estimated numbers of boat trips, by month and subarea, 2018. Upper panel: estimated monthly non-lodge boat trips. Lower panel: total effort, calculated as the sum of lodge and non-lodge boat trips.

to 7.2 trips per day per subarea (in the June 16- July 9 period), and peaking at 9.0 trips per day per subarea (from July 10-31), but the differences were not statistically significant ($\chi^2 = 4.0$, $df = 3$, $P = 0.26$).

Total lodge effort in 2018 was 577, 576, and 250 boat trips in June, July and August, respectively (Table 13). To calculate total (lodge + non-lodge) fishing effort by subarea, we

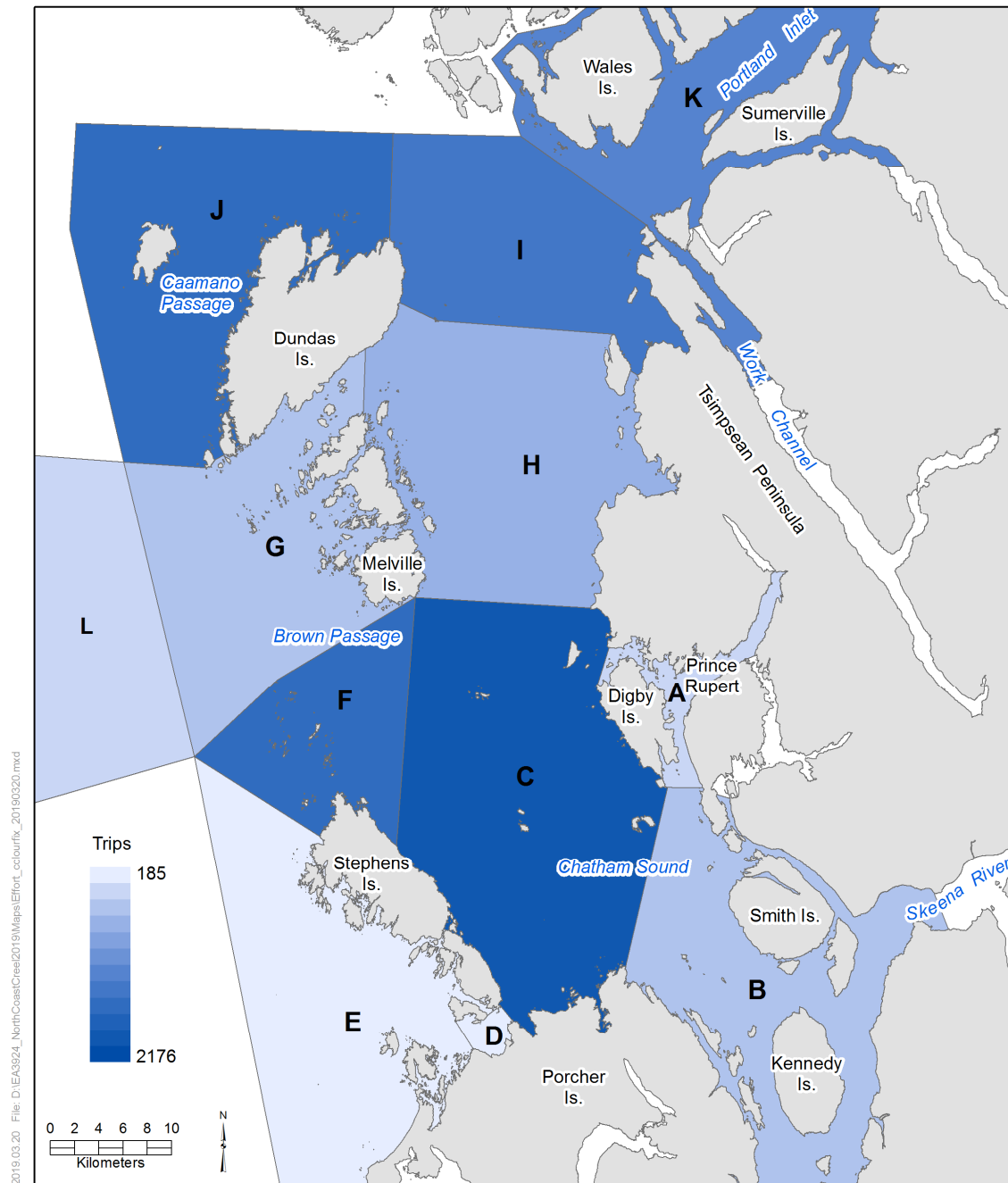


Figure 8. Heat map showing the estimated total effort (numbers of boat trips, lodge + non-lodge) in each subarea in 2018. See Table 13 for details.

distributed the lodge boat trips among the three subareas in Management Area 3, according to the relative non-lodge effort in each (Table 13; Figure 7). In all, there were 2,873 (SE = 13) boat trips in June, 4,593 (SE = 17) boat trips in July, and 3,304 (SE = 13) boat trips in August. The full-season Area 3/4 effort estimate for 2018 was 10,770 boat trips (SE = 25).

May Effort

Trailer counts began in May of 2018, and ran until the end of the creel monitoring period (Figure 9). A visual examination of the season's worth of trailer counts revealed, to our eyes, four phases: 1) a phase of relatively consistent effort from May 1 to June 8; 2) a phase featuring elevated activity from June 9 to July 8; 3) a phase with even higher activity from July 9 to August 6; and 4) a phase during which effort markedly dropped off from August 7 onward (Figure 9). To these four phases we assigned the most-appropriate angler activity patterns: 1) that from the June 1-15 period; 2) that from the June 16 to July 9 period; 3) that from the July 10-31 period; and 4) that for the August 1-31 period.

The regression line from Figure 6 was used to calculate what the daily boat counts *would have been* had there been overflights each day, and the day-type and phase-specific activity pattern data (Figure 4) were used to expand the boat counts to calculate total fishery effort by day (**Error! Reference source not found.**). Summing over months revealed that total effort in May 018 was fairly significant, equivalent to 72% of that estimated for June (about 2,068 trips).

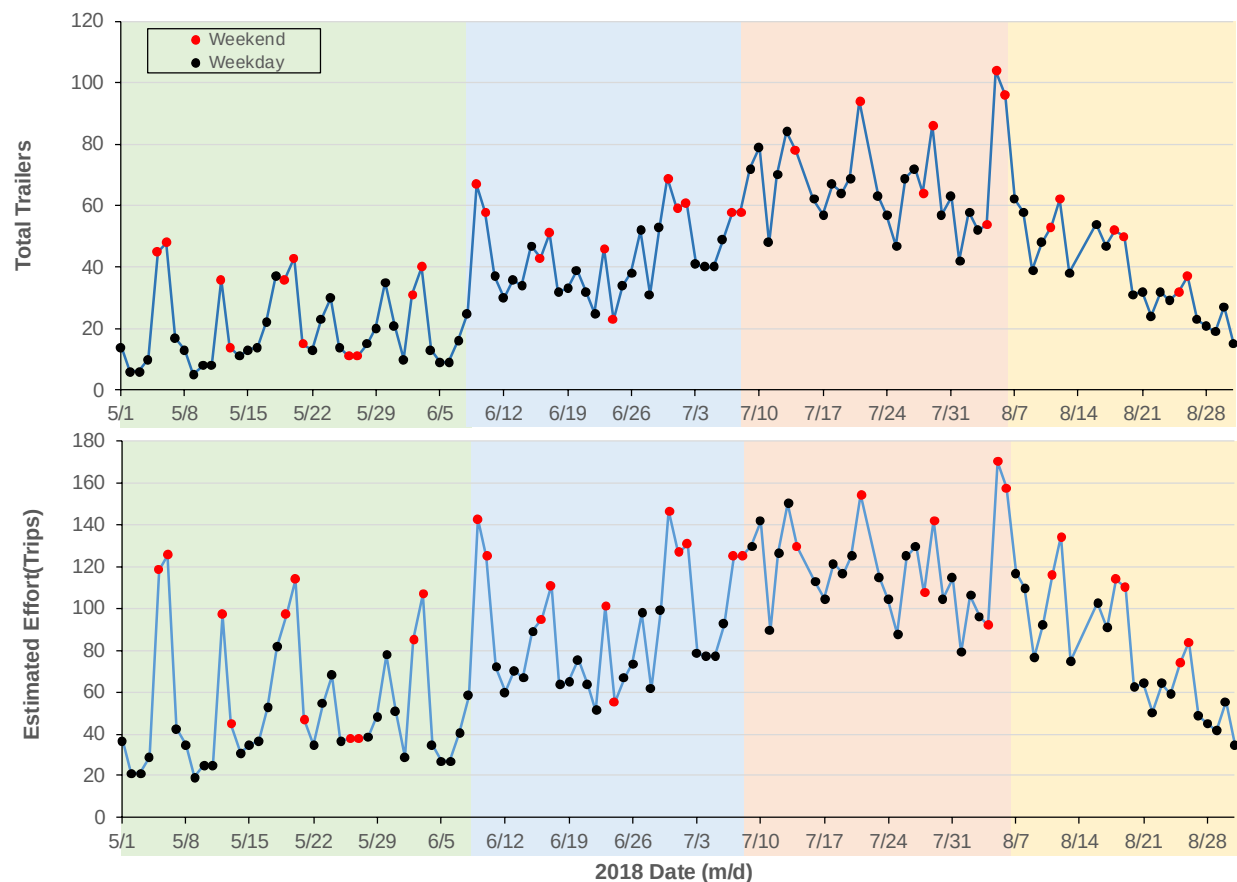


Figure 9. Total number of trailers counted at landing sites (upper), and total fishery effort (lower), by date, May through August. Upper: a visual examination revealed, to our eyes, four phases: 1) a phase of relatively consistent effort from May 1 to June 8; 2) a phase featuring elevated activity from June 9 to July 8; 3) a phase with even higher activity from July 9 to August 6; and 4) a phase during which effort markedly dropped off from August 7 onward. Lower: estimates made use of boat-count-from-trailer-count regression (Figure 6) and angler activity profiles (Figure 4).

Catch Estimates

Monthly and full-season estimates were generated of the numbers of fish released and retained, by species (Tables 14-23; Appendix 4). Non-lodge estimates were generated from the product of effort and CPE estimates. Lodge catches were extracted from census documents. The results presented in this section are total (lodge + non-lodge) numbers of fish retained and released.

An estimated 5,830 (SE = 198) Chinook Salmon were retained by anglers in Areas 3 or 4 during 2018 the study period (Table 15; Figure 10). The largest proportional harvest came from subarea 4F (21%), followed by 3J (20%), 3I (18%) and 3K (15%). Harvest differed markedly over the study period: it was significantly lower from June 16 to July 9 than in any other period ($\chi^2 = 21.6$, $df = 3$, $P < 0.0001$; Table 24). During the study, the number of released Chinook Salmon was estimated at 5,997 (SE = 287), or 51% of the total catch. During the period from June 16 to July 9, retention (calculated for non-lodge anglers) was 3%, in contrast to 55-61% during the other three periods.

The salmonid that was harvested in the greatest numbers during the 2018 study period was, by far, Coho Salmon (10,464 fish retained, SE = 434; Table 16; Figure 11). The largest proportional harvest came from subarea 4F (24%), followed by subareas 3I, 3J, and 3K (15-20%). Harvest differed markedly over the study period: that from June 1-15 was significantly lower than all other periods, harvest from June 16-July 9 was significantly lower than the following two periods, and the last two periods did not differ significantly ($\chi^2 = 27.3$, $df = 3$, $P < 0.0010$; Table 24). During the study period, an estimated 567 (SE = 147) Coho Salmon were released by anglers (5% of the total catch).

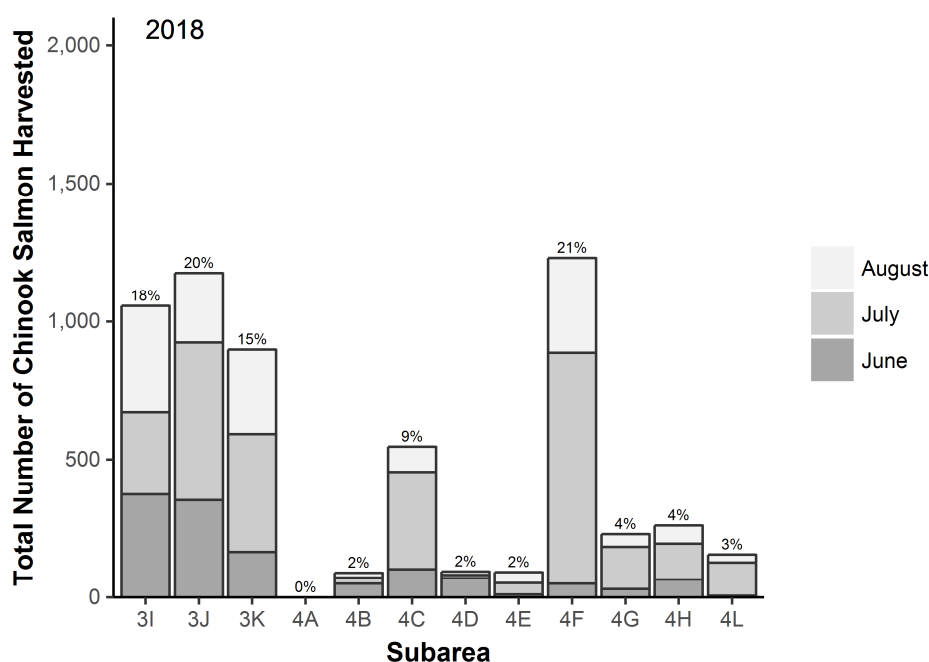


Figure 10. Total (lodge + non-lodge) estimated Chinook Salmon harvest, by month and subarea, 2018.

Table 14. Total (lodge + non-lodge) estimated harvest, by subarea, for popular sportfish taxa, 2018.

Species	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
Chinook Salmon	1,056	1,176	899	1	89	546	94	91	1,231	230	261	156	3,131	2,700	5,830	154	125	198
Coho Salmon	1,531	1,960	2,095	0	37	404	68	288	2,459	393	331	897	5,586	4,878	10,464	325	287	434
Pink Salmon	128	310	306	0	11	136	2	18	349	14	74	45	744	650	1,394	146	61	158
Chum Salmon	82	20	31	0	0	7	2	1	15	7	7	4	133	44	177	44	10	45
Sockeye Salmon	0	8	24	0	0	0	0	0	0	0	0	0	32	0	32	14	0	14
Unknown Salmon	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Pacific Halibut	1,411	1,921	585	0	108	1,519	180	327	2,060	663	947	1,133	3,916	6,936	10,853	218	195	293
Lingcod	192	1,858	138	5	5	124	83	169	849	211	248	388	2,188	2,082	4,270	161	126	205
other groundfish *	128	106	29	35	34	175	24	14	427	23	176	16	263	925	1,187	47	91	103
Rockfish †	417	1,973	550	9	118	984	302	293	1,498	420	600	737	2,941	4,962	7,903	282	224	360

* includes Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and unknown groundfish

† includes Black, Bocaccio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail and unknown rockfish

Table 15. Total (lodge + non-lodge) estimated Chinook Salmon catch, by month and subarea, shown separately for released and retained fish, 2018.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	373.7	352.1	164.4	0.6	50.1	102.1	69.1	10.1	50.0	29.9	65.1	6.1	890	383	1,273	52	36	63
Released	739.5	421.3	224.6	0.3	131.1	92.8	68.1	36.0	251.2	6.9	104.4	44.4	1,385	735	2,121	115	82	141
July																		
Retained	297.0	571.9	427.4	0	21.9	353.0	11.7	41.9	837.3	153.1	129.4	119.9	1,296	1,668	2,964	115	108	158
Released	353.5	376.6	379.5	0.2	12.8	480.3	27.6	20.1	593.8	131.8	146.8	89.2	1,110	1,503	2,612	125	154	198
August																		
Retained	385.7	251.7	306.9	0	17.5	91.4	13.3	39.5	343.4	46.5	66.9	29.8	944	648	1,593	88	51	102
Released	357.9	129.8	501.9	0	8.7	84.4	3.3	15.7	144.6	5.4	0	12.8	990	275	1,264	147	40	152
TOTAL																		
Retained	1,056	1,176	899	1	89	546	94	91	1,231	230	261	156	3,131	2,700	5,830	154	125	198
Released	1,451	928	1,106	1	153	657	99	72	990	144	251	146	3,484	2,513	5,997	225	179	287
Catch	2,507	2,103	2,005	1	242	1,204	193	163	2,220	374	513	302	6,615	5,212	11,828	272	218	349

Table 16. Total (lodge + non-lodge) estimated Coho Salmon catch, by month and subarea, shown separately for released and retained fish, 2018.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	23.5	0	13.8	0	0	17.6	0	12.7	38.8	0	4.1	15.3	37	89	126	14	18	22
Released	0	0	0	0	0	0	7.0	0	2.9	0	2.0	0	0	12	12	0	7	7
July																		
Retained	565.9	1129.8	597.5	0	13.9	210.6	6.5	191.8	1833.1	269.4	210.4	728.6	2,293	3,464	5,758	222	276	354
Released	160.9	88.5	0	0	2.0	8.7	5.0	0	111.0	0	1.2	23.2	249	151	401	136	41	143
August																		
Retained	941.6	830.4	1483.5	0	23.3	175.8	61.9	83.5	587.4	123.4	116.2	153.3	3,256	1,325	4,580	238	80	251
Released	11.4	0	84.9	0	0	0	16.6	9.9	31.6	0	0	0	96	58	154	32	17	36
TOTAL																		
Retained	1,531	1,960	2,095	0	37	404	68	288	2,459	393	331	897	5,586	4,878	10,464	325	287	434
Released	172	89	85	0	2	9	29	10	146	0	3	23	346	221	567	140	45	147
Catch	1,703	2,049	2,180	0	39	413	97	298	2,605	393	334	920	5,932	5,099	11,030	354	291	459

Table 17. Total (lodge + non-lodge) estimated Pink Salmon catch, by month and subarea, shown separately for released and retained fish, 2018.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	12.8	7.4	7.7	0	0	3.6	0	0	5.8	0	0	0	28	9	37	9	4	10
Released	0	0	0	0	0	3.6	0	0	0	0	0	0	0	4	4	0	3	3
July																		
Retained	115.2	248.3	68.2	0	7.9	104.3	1.3	8.4	253.2	0	73.9	15.5	432	465	896	112	57	126
Released	107.3	29.5	88.2	0	2.0	52.1	0	0	104.0	85.1	23.1	0	225	266	491	90	81	121
August																		
Retained	0	54.1	230.4	0	2.9	28.1	1.1	9.4	90.4	14.3	0	29.8	285	176	461	93	23	96
Released	0	30.9	24.2	0	5.8	28.1	0	0.9	9.0	0	0	0	55	44	99	18	17	25
TOTAL																		
Retained	128	310	306	0	11	136	2	18	349	14	74	45	744	650	1,394	146	61	158
Released	107	60	112	0	8	84	0	1	113	85	23	0	280	314	594	92	83	124
Catch	235	370	419	0	19	220	2	19	462	99	97	45	1,024	964	1,988	172	103	201

Table 18. Total (lodge + non-lodge) estimated Chum Salmon catch, by month and subarea, shown separately for released and retained fish, 2018.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	0	0	0	0	0	4.4	0	0	0	0	0	0	0	4	4	0	3	3
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
July																		
Retained	68.0	0	0	0	0	2.7	0	0	10.4	0	0	0	68	13	81	39	6	40
Released	0	0	0	0	0	0	0	0	26.0	0	0	0	0	26	26	0	16	16
August																		
Retained	14.5	19.6	30.7	0	0	0	2.2	1.3	4.5	7.2	7.0	4.3	65	27	91	19	7	20
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL																		
Retained	82	20	31	0	0	7	2	1	15	7	7	4	133	44	177	44	10	45
Released	0	0	0	0	0	0	0	0	26	0	0	0	0	26	26	0	16	16
Catch	82	20	31	0	0	7	2	1	41	7	7	4	133	70	203	44	19	48

Table 19. Total (lodge + non-lodge) estimated Sockeye Salmon catch, by month and subarea, shown separately for released and retained fish, 2018.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
July																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
August																		
Retained	0	7.7	24.2	0	0	0	0	0	0	0	0	0	32	0	32	14	0	14
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL																		
Retained	0	8	24	0	0	0	0	0	0	0	0	0	32	0	32	14	0	14
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Catch	0	8	24	0	0	0	0	0	0	0	0	0	32	0	32	14	0	14

Table 20. Total (lodge + non-lodge) estimated Pacific Halibut catch, by month and subarea, shown separately for released and retained fish, 2018.

Month / Catch Type		Subarea											TOTALS			SE of the Totals			
		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																			
	Retained	485.0	744.7	90.7	0	25.3	396.4	82.4	98.2	488.8	266.9	380.6	347.7	1,320	2,086	3,407	101	92	137
	Released	223.4	381.3	51.9	0	23.8	278.7	50.7	48.9	303.5	147.3	352.0	484.1	657	1,689	2,345	91	120	150
July																			
	Retained	644.5	581.4	208.0	0	33.1	608.9	56.5	144.9	893.1	193.8	432.8	465.9	1,434	2,829	4,263	161	142	214
	Released	563.4	579.7	147.2	0	15.4	396.1	27.5	117.3	452.4	90.5	377.2	623.0	1,290	2,099	3,390	320	177	366
August																			
	Retained	281.3	594.9	286.0	0	49.5	513.4	40.9	83.9	677.8	202.1	133.8	319.4	1,162	2,021	3,183	107	98	145
	Released	297.1	618.7	363.7	0	8.7	168.8	19.9	42.2	343.4	178.9	49.3	366.3	1,280	1,177	2,457	263	100	281
TOTAL																			
	Retained	1,411	1,921	585	0	108	1,519	180	327	2,060	663	947	1,133	3,916	6,936	10,853	218	195	293
	Released	1,084	1,580	563	0	48	844	98	208	1,099	417	778	1,473	3,226	4,966	8,192	424	236	485
	Catch	2,495	3,501	1,147	0	156	2,362	278	535	3,159	1,080	1,726	2,606	7,143	11,902	19,045	477	306	567

Table 21. Total (lodge + non-lodge) estimated Lingcod catch, by month and subarea, shown separately for released and retained fish, 2018.

		Subarea											TOTALS			SE of the Totals			
Month / Catch Type		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																			
	Retained	172.1	663.9	116.6	0	2.8	24.0	68.9	53.7	251.1	98.9	136.1	165.4	953	801	1,754	98	75	123
	Released	24.7	102.3	0	0	5.6	11.6	5.3	59.6	37.9	57.5	26.0	25.3	127	229	356	46	46	65
July																			
	Retained	7.6	860.3	8.4	0	2.2	86.3	8.1	94.6	412.8	55.1	76.4	128.7	876	864	1,740	113	87	142
	Released	13.4	130.9	0	0	6.0	28.8	36.4	51.9	20.1	29.7	19.2	11.7	144	204	348	50	45	67
August																			
	Retained	12.3	333.5	13.1	5.5	0	14.1	5.5	20.6	185.3	57.2	35.2	93.7	359	417	776	61	52	80
	Released	0	23.2	0	0	0	0	1.1	2.2	18.1	8.9	0	21.3	23	52	75	13	17	21
TOTAL																			
	Retained	192	1,858	138	5	5	124	83	169	849	211	248	388	2,188	2,082	4,270	161	126	205
	Released	38	256	0	0	12	40	43	114	76	96	45	58	295	484	779	69	66	96
	Catch	230	2,114	138	5	17	165	125	283	925	307	293	446	2,482	2,566	5,049	175	143	226

Table 22. Total (lodge + non-lodge) estimated catch of other groundfish, by month and subarea, shown separately for released and retained fish, 2018. Included are Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and other miscellaneous groundfishes.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	62.2	36.9	26.7	0	11.1	78.8	8.8	3.0	83.6	13.8	90.2	10.0	126	299	425	30	47	56
Released	71.7	94.9	60.8	0	53.3	237.7	159.6	6.4	66.1	0	17.5	18.4	227	559	786	56	148	159
July																		
Retained	31.4	53.3	2.5	23.9	7.9	61.1	4.4	0	276.0	7.1	82.7	6.2	87	469	557	29	74	79
Released	17.9	52.4	32.2	41.4	107.3	190.7	2.6	0	217.0	0	22.5	3.9	103	585	688	34	92	98
August																		
Retained	34.3	15.5	0	10.9	14.6	35.2	11.1	11.2	67.8	1.8	3.5	0	50	156	206	22	27	35
Released	114.3	108.3	97.0	20.8	64.0	133.6	37.6	35.0	198.8	109.1	197.1	21.3	320	817	1,137	90	94	130
TOTAL																		
Retained	128	106	29	35	34	175	24	14	427	23	176	16	263	925	1,187	47	91	103
Released	204	256	190	62	225	562	200	41	482	109	237	44	649	1,962	2,611	111	198	227
Catch	332	361	219	97	258	737	224	56	909	132	414	60	912	2,886	3,799	121	218	250

Table 23. Total (lodge + non-lodge) estimated rockfish catch, by month and subarea, shown separately for released and retained fish, 2018. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	243.4	453.3	144.4	2.3	28.2	262.0	202.7	84.2	434.6	184.1	309.5	290.3	841	1,798	2,639	86	126	152
Released	192.7	467.1	154.1	0	212.8	365.9	519.0	161.0	490.0	437.2	568.2	710.8	814	3,465	4,279	125	275	302
July																		
Retained	113.2	1098.6	198.6	1.0	87.3	518.2	57.3	149.1	634.2	96.2	255.4	178.8	1,410	1,978	3,388	248	151	290
Released	116.4	1009.5	280.5	6.4	369.2	1300.5	244.0	691.9	1186.5	243.3	483.3	493.7	1,406	5,019	6,425	243	516	570
August																		
Retained	60.8	421.4	207.1	5.5	2.9	203.9	42.0	59.7	429.3	139.5	35.2	268.3	689	1,186	1,876	103	107	148
Released	217.1	835.2	254.6	10.9	43.7	717.3	55.3	88.9	659.7	121.6	45.8	221.5	1,307	1,965	3,272	272	186	329
TOTAL																		
Retained	417	1,973	550	9	118	984	302	293	1,498	420	600	737	2,941	4,962	7,903	282	224	360
Released	526	2,312	689	17	626	2,384	818	942	2,336	802	1,097	1,426	3,527	10,448	13,975	385	613	724
Catch	944	4,285	1,239	26	744	3,368	1,120	1,235	3,834	1,222	1,697	2,163	6,468	15,410	21,878	477	653	809

Table 24. Kruskal Wallis tests for differences in harvest (fish caught per boat trip per subarea) among months (left) and periods (right), by species, 2018. Subareas treated as replicates.

Species	EFFECT OF MONTH						EFFECT OF PERIOD						
	Kruskal-Wallis Statistics			Median Monthly Harvest			Kruskal-Wallis Statistics			Median Harvest by Period			
	χ^2	df	P	June	July	August	χ^2	df	P	June 1-15	June 16 - July 9	July 10-31	August
Chinook Salmon	2.2	2	0.33	57.6	141.2	56.7	21.6	3	<0.0001	57.6	0	139.1	56.7
Chum Salmon	11.1	2	0.004	0	0	4.4	19.7	3	0.0002	0	0	0	4.4
Coho Salmon	14.8	2	0.0006	7.3	240.0	138.4	27.3	3	<0.0001	0	12.2	238.7	138.4
Pink Salmon	9.1	2	0.011	0	40.8	11.9	17.3	3	0.0006	0	0	40.8	11.9
Sockeye Salmon	4.1	2	0.13	0	0	0	6.1	3	0.11	0	0	0	0
Arrow Flounder	2.2	2	0.33	0	0	0	2.0	3	0.56	0	0	0	0
Big Skate	4.1	2	0.13	0	0	0	6.1	3	0.11	0	0	0	0
Cabezon	3.6	2	0.17	0	0	0	6.1	3	0.11	0	0	0	0
English Sole	0.4	2	0.83	0	0	0	2.1	3	0.55	0	0	0	0
Greenling	0.3	2	0.88	0	0	0	0.6	3	0.89	0	0	0	0
Lingcod	1.7	2	0.43	83.9	65.8	17.4	2.5	3	0.47	23.1	47.4	12.9	17.4
Pacific Cod	3.0	2	0.22	11.8	6.7	3.6	1.5	3	0.68	3.7	5.7	1.9	3.6
Pacific Halibut	0.5	2	0.80	307.3	313.3	238.2	9.4	3	0.024	45.9	405.6	198.9	238.2
Ratfish	0.0	2	1	0	0	0	0.0	3	1	0	0	0	0
Rock Sole	2.0	2	0.37	0	0	0	2.0	3	0.56	0	0	0	0
Sablefish	0.0	2	0.99	0.1	1.2	0.6	4.1	3	0.25	0	0	0	0.6
Starry Flounder	0.0	2	1	0	0	0	0.0	3	1	0	0	0	0
Black Rockfish	1.5	2	0.46	11.3	5.1	12.7	6.4	3	0.10	0	2.7	0	12.7
Boccacio Rockfish	0.0	2	1	0	0	0	0.0	3	1	0	0	0	0
Canary Rockfish	4.1	2	0.13	0	0	0	6.1	3	0.11	0	0	0	0
China Rockfish	0.8	2	0.69	0	0	0	4.7	3	0.20	0	0	0	0
Copper Rockfish	0.2	2	0.90	0	0	0.9	2.5	3	0.48	0	0	0	0.9
Dusky Rockfish	0.3	2	0.85	0	0	0	2.0	3	0.58	0	0	0	0
Quillback Rockfish	0.5	2	0.79	17.6	19.4	13.8	8.0	3	0.047	0.2	22.9	9.3	13.8
Redband Rockfish	2.0	2	0.37	0	0	0	3.0	3	0.39	0	0	0	0
Redstripe Rockfish	2.0	2	0.37	0	0	0	3.0	3	0.39	0	0	0	0
Tiger Rockfish	3.3	2	0.19	0	0	0.4	9.3	3	0.026	0	0	0	0.4
Widow Rockfish	0.0	2	1	0	0	0	0.0	3	1	0	0	0	0
Yelloweye Rockfish	2.0	2	0.37	0	0	0	2.6	3	0.46	0	0	0	0
Yellowtail Rockfish	1.2	2	0.56	0	0	0	3.4	3	0.33	0	0	0	0

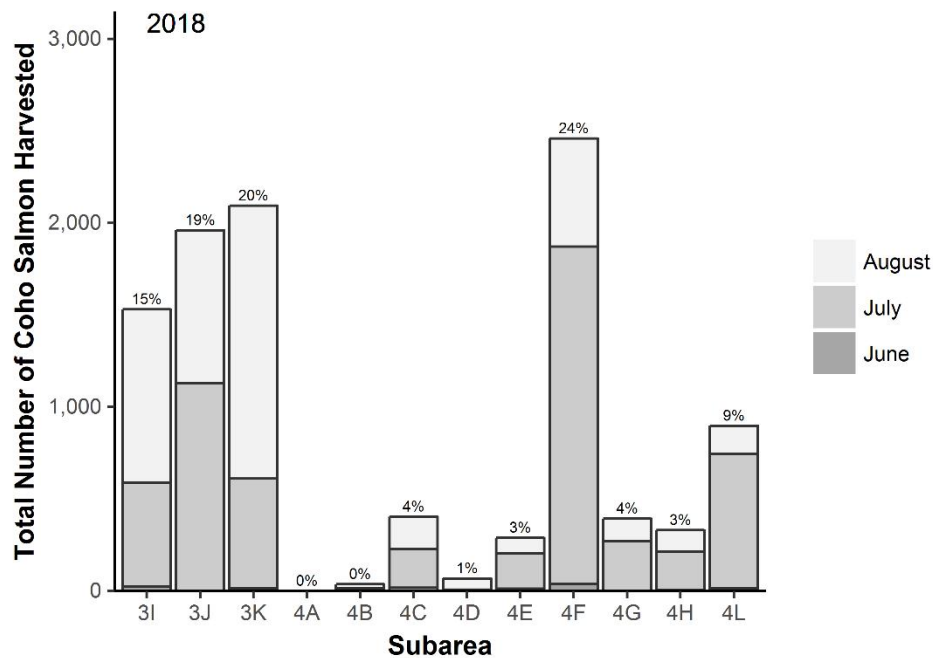


Figure 11. Total (lodge + non-lodge) estimated Coho Salmon harvest, by month and subarea, 2018.

Pink salmon was the least likely salmonid to be retained in 2018 (Table 17). We estimate that 1,988 (SE = 201) fish were caught, of which 70% were retained (1,394 fish, SE = 158), and 30% were released (594 fish, SE = 124). Harvest of Pink Salmon varied significantly among periods ($\chi^2 = 17.3$, df = 3, $P < 0.0006$; Table 24), with all significantly lower harvest at the start of the survey (June 1 - July 9) compared to the end (July 10 - August 31). Very few Chum (177 fish, SE = 45) or Sockeye (32 fish, SE = 14) salmon were harvested in Area 3 and 4 during the study period (Tables 18 and 19).

An estimated 10,853 (SE = 293) Pacific Halibut were harvested in Areas 3 or 4 during the 2018 study period (Table 20; Figure 12). Most of the harvest could be attributed to effort in subareas 3I, 3J, 4C, and 4F (together 64%). Harvest varied among periods, with significantly fewer fish harvested in the June 1-15 period, as compared to June 16-July 9, or in August ($\chi^2 = 9.4$, df = 3, $P = 0.024$; Table 24). We estimated that 8,192 (SE = 485) Pacific Halibut were released (43% of the total catch).

From June to August 2018, an estimated 4,270 (SE = 193) Lingcod were harvested, with higher median harvest in the June 16 – July 9 period than in other periods, though differences were not statistically significant. ($\chi^2 = 2.5$, df = 3, $P = 0.47$; Table 21; Figure 13). The harvest occurred largely in subarea 3J (44%), and F (20%). Of the captured Lingcod, 15% were released (779 fish, SE = 96).

An estimated 3,799 (SE = 250) other ground fish were caught June to August 2018, of which 69% was released after capture. During the study period, an estimated 1,187 (SE = 103) individuals were retained (Table 22), mostly (38%) Sablefish (*Anoplopoma fimbria*), but also

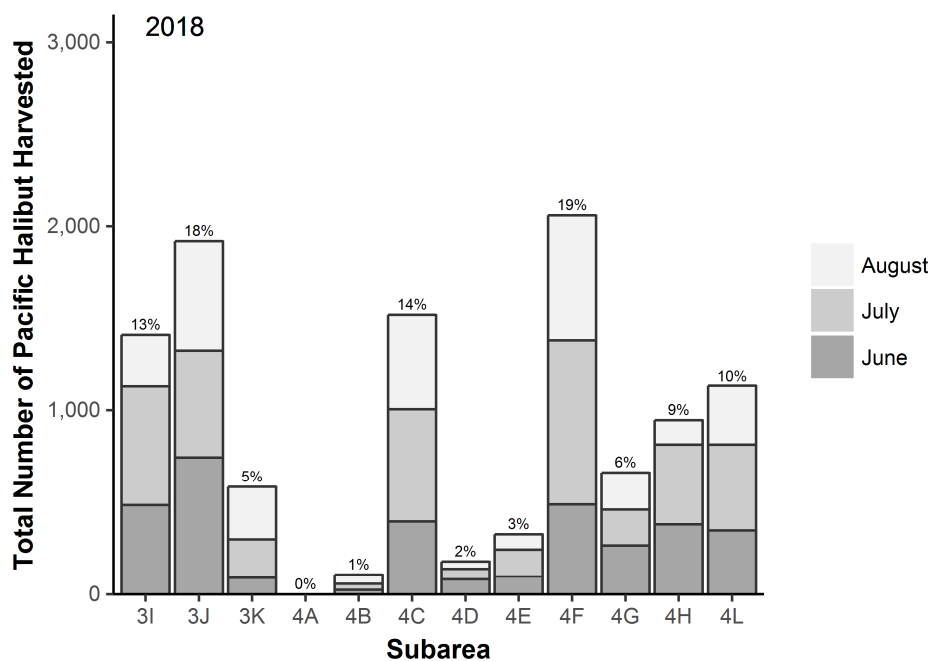


Figure 12. Total (lodge + non-lodge) estimated Pacific Halibut harvest, by month and subarea, 2018.

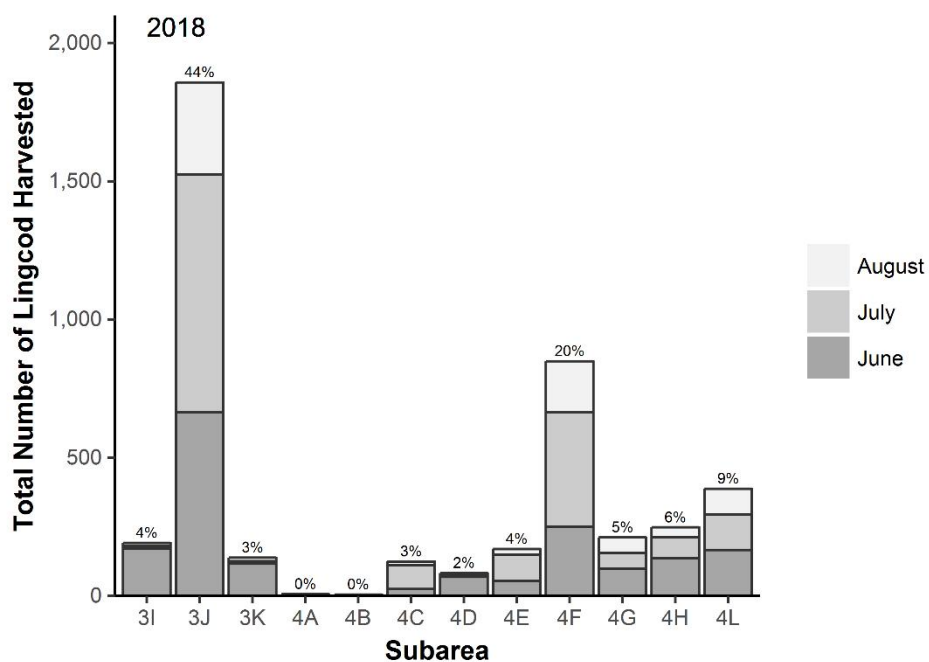


Figure 13. Total (lodge + non-lodge) estimated Lingcod harvest, by month and subarea, 2018.

including Pacific cod (*Gadus macrocephalus*, 37%), greenlings (Hexagrammidae, 14%), Cabezon (*Scorpaenichthys marmoratus*), and various species of skates, flounders, and soles.

Rockfish harvest was estimated at 7,903 fish (SE = 360; Table 23), composed mostly of unknown (68%), Quillback (*Sebastes maliger*, 17%), and Black (*S. melanops*, 8%), Copper (*S. caurinus*, 2%), China (*S. nebulosus*, 2%), and seven other rockfish species. In all, 13,975 (SE = 724) were released. Various rockfishes showed significant harvest differences among periods (Table 24), including Quillback (harvest June 1-15 was significantly lower than June 16 – July 9) and Tiger (harvest July 10-31 was significantly less than in August) rockfish.

Biosampling

During interviews, observers were told about 733 retained Chinook Salmon, and were permitted to biosample 285 fish. The observed sex ratio was 56:44 (male:female, $n = 117$), and most fish were red-fleshed (91%, $n = 159$; Table 25). The median fork length of biosampled Chinook Salmon ($n = 227$) was 68 cm (95% of observations were between 51 and 100 cm; Figure 14). In all, 47 adipose-clipped individuals were encountered (Table 25), including 36 from standardized sampling methods (anglers were asked if their catch could be biosampled before any other interactions took place; Appendix 5) and 11 from non-standard interactions. Of the 47, 79% had their heads collected and labelled for submission to the Salmon Head Recovery Program (Table 25). Of the standardized samples, 15% were adipose-clipped (Appendix 5). Scale samples were obtained from 80% of the biosampled Chinook Salmon (Table 25).

For Coho Salmon, observers were told about 1,334 retained fish, and were permitted to biosample 346 fish. The observed sex ratio was 36:64 (male:female; $n = 132$; Table 25). The median Coho Salmon fork length ($n = 298$) was 67 cm (95% of observations were between 52 and 78 cm; Figure 14). In all, 5 adipose-clipped individuals were encountered (all while using standardized sampling methods), of which 80% had their heads collected and labelled for submission to the Salmon Head Recovery Program (Table 25). Of the standardized samples, 1.6% were adipose-clipped (Appendix 5). Scale samples were obtained from 86% of the biosampled Coho Salmon (Table 25).

Of the 1,741 Pacific Halibut that anglers reported as retained, 543 were biosampled (Table 25). Only 28 fish were sexed, and 542 were measured. The median size was 82 cm (95% of observations were between 56 and 114 cm; Figure 14). Average net and round weights, calculated from average lengths using regression formulas, were 6.0 kg (13.1 lb) and 7.9 kg (17.4 lb) in Area 3, and 5.6 kg (12.4 lb) and 7.4 kg (16.4 lb) in Area 4, respectively (Table 26).

Harvested Halibut Biomass

Based on the average weights of biosampled Pacific Halibut, and the estimated harvest, it was estimated that total harvested Pacific Halibut net biomass was 62.3 tonnes (137,249 lb), including 23.3 t (51,504 lb) in Area 3 and 38.9 t (85,843 lb) in Area 4. In terms of round weight, Pacific Halibut harvest was 82.4 t (181,769 lb) including 30.9 t (68,080 lb) in Area 3 and 51.6 t (113,689 lb) in Area 4 (Table 26).

Table 25. Biosampling results, 2018.

	Chinook Salmon		Coho Salmon		Pacific Halibut	
	Number	Percent	Number	Percent	Number	Percent
Number Examined	285		346		543	
Subarea						
3I	19	7%	17	5%	27	5%
3J	23	8%	12	3%	23	4%
3K	19	7%	30	9%	4	1%
4A	3	1%				
4B	20	7%	8	2%	28	5%
4C	35	12%	8	2%	97	18%
4D	17	6%	17	5%	45	8%
4E	2	1%	21	6%	19	4%
4F	108	38%	145	42%	133	25%
4G	9	3%	11	3%	27	5%
4H	20	7%	24	7%	68	13%
4L	7	2%	51	15%	60	11%
Sex						
Male	66	56%	47	36%	11	39%
Female	51	44%	85	64%	17	61%
Undeterminable	145		180		301	
Unknown	23		34		214	
Trip Type						
Guided	115	40%	140	40%	224	41%
Unguided	170	60%	206	60%	319	59%
Adipose Clips						
Clipped	47	see	5	see		
Unmarked	232	App 5	336	App 5		
Unknown	1		1			
Head Tagged						
Yes	37	13%	4	1%		
No	248	87%	342	99%		
Scale Samples						
Yes	227	80%	299	86%		
No	58	20%	47	14%		
Flesh Colour						
Red	144	91%				
White	15	9%				
Marble						
Unknown						

Table 26. Average weight, and harvest (numbers and biomass) of Pacific Halibut in Area 3 and 4, 2018.

	Stat Area 3				Stat Area 4			
	Net Weight		Round Weight		Net Weight		Round Weight	
	kg	lb	kg	lb	kg	lb	kg	lb
Creel Interview (Non-Lodge) Average Weights								
Guided	6.1	13.5	8.1	17.8	5.7	12.5	7.5	16.6
Unguided	5.6	12.4	7.5	16.5	5.6	12.3	7.4	16.3
Overall	6.0	13.1	7.9	17.4	5.6	12.4	7.4	16.4
Estimated Non-Lodge Harvest (number of fish)								
Overall		3,514				6,936		
Estimated Lodge Harvest (number of fish)								
Overall		402				0		
Biomass Harvested								
Non-lodge	20,924	46,129	27,711	61,092	38,938	85,843	51,568	113,689
Lodge	2,393	5,276	3,170	6,988	0	0	0	0
Overall	23,317	51,405	30,881	68,080	38,938	85,843	51,568	113,689

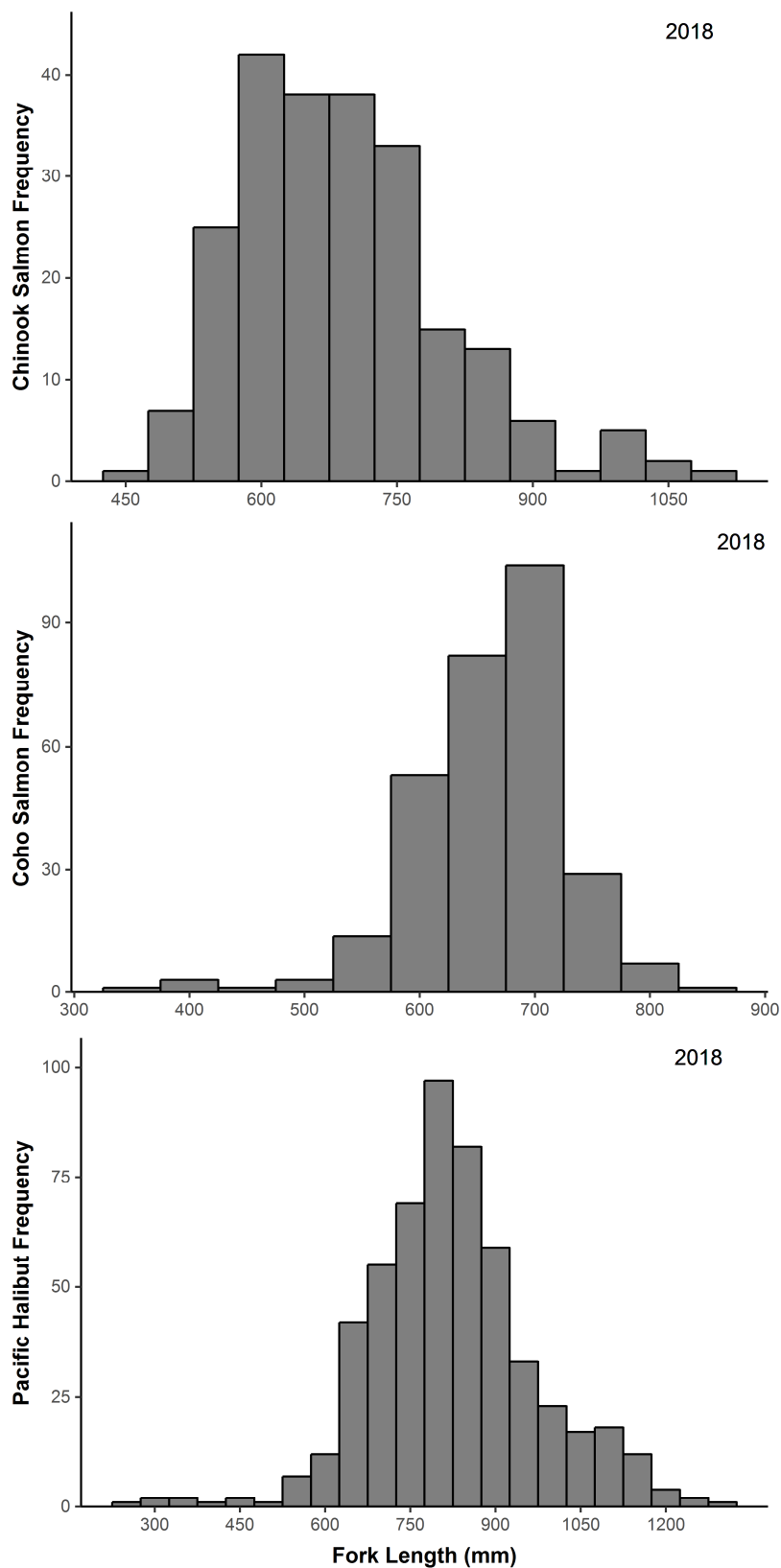


Figure 14. Length-frequency distributions of biosampled Chinook (n = 227) and Coho (n = 298) salmon, and Pacific Halibut (n = 542), 2018. Note that X and Y axis scales vary between plots.

DISCUSSION

Creel Estimates

In general, recreational fishing effort and salmon catch in Areas 3 & 4 have been increasing over the last 20 years (Tables 27 and 28). Using published creel data collected since 1995, we found ten years (1995-96, 2001-02, 2008-09¹, and 2015-2018) for which effort data were collected in June, July and August (Van Tongeren and Winther 2010, Robichaud et al. 2016, 2017, Robichaud and Addison 2018), and we have therefore limited analyses to those three months. Meta-analysis revealed (Figure 15) that fishing effort has increased steadily (by 296 trips per year on average) since 1995. Effort in 2018 was lower relative to other recent years (2015-2017), but looking at the long-term trend, the 2018 value still clusters with other recent years and supports the notion that effort gradually increases over time.

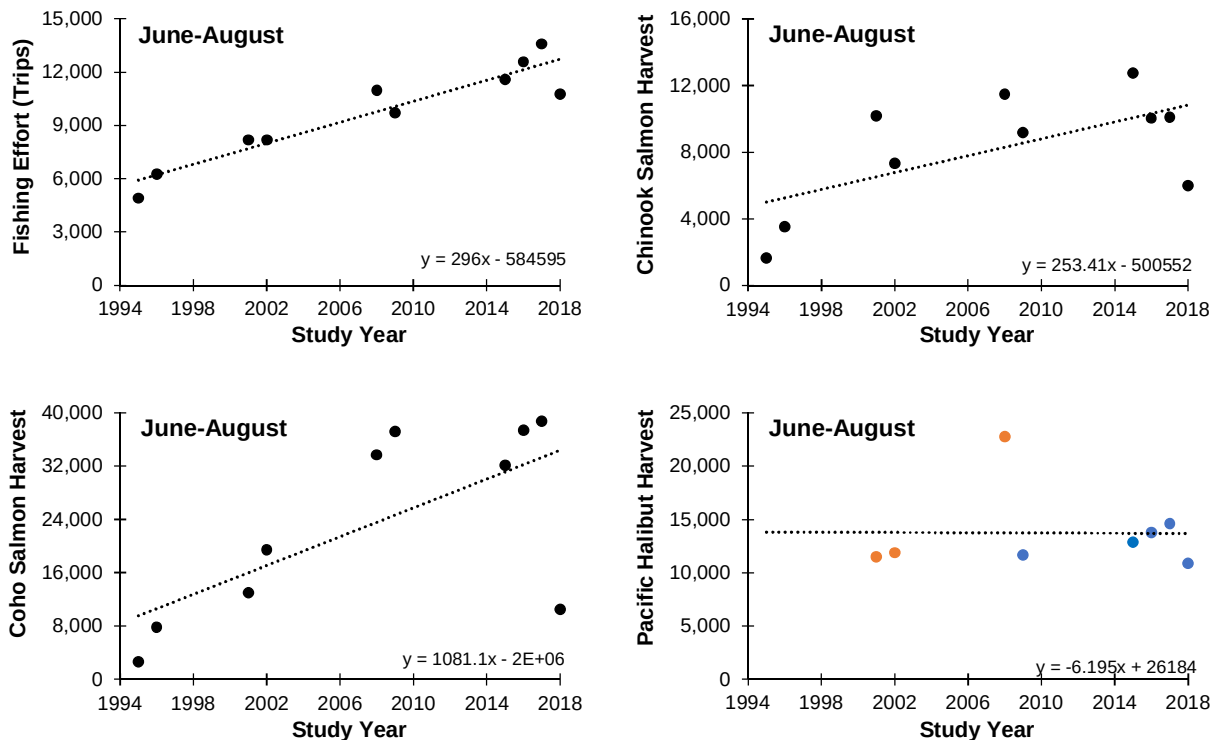


Figure 15. Fishing effort, and harvest of Chinook and Coho salmon, and Pacific Halibut from 1995 to 2018. Annual estimates only include data from June, July and August. Only years with data from all three months are included. Linear regression equations are shown in the lower right of each panel. For Pacific Halibut, bag limits changed after 2008 from two fish per day (orange dots) to one fish per day (blue dots).

¹ Creel data from 2010-2014 were collected and analyzed by DFO, and preliminary results were presented as part of their post-season review process. We based our meta-analysis on published accounts (written accounts showing methods, results, and discussing the caveats of the results), and hence have not included data from these years.

Table 27. Monthly recreational fishing effort (boat trips) in Area 3 and 4 from 1995 to 2018, as estimated using creel survey methods.

	Year																					
Month	1995	1996	1997	1998	1999	2000	2001	2002	2003	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
May	965	391	--	--	--	--	449	960	--	--	--	983	--	--	--	--	--	--	--	--	--	--
June	1,590	2,490	--	--	--	--	3,801	3,463	--	--	--	4,696	3,043	--	--	--	--	--	3,424	3,961	4,574	2,873
July	1,394	2,385	--	1,307	--	1,842	2,588	2,892	--	--	--	3,750	3,804	--	--	--	--	--	4,279	5,295	6,044	4,593
Aug	1,927	1,400	--	1,351	--	1,327	1,808	1,843	--	2,593	--	2,529	2,865	--	--	--	--	--	3,881	3,325	2,980	3,304
Sept	864	488	--	504	--	954	540	676	--	758	--	394	817	--	--	--	--	--	--	--	--	--
TOTAL	6,740	7,154	--	3,162	--	4,123	9,186	9,834	--	3,351	--	12,352	10,529	--	--	--	--	--	11,585	12,581	13,598	10,770

Table 28. Monthly Chinook Salmon, Coho Salmon and Pacific Halibut harvest in the Area 3 and 4 recreational fishery from 1995 to 2018, as estimated using creel survey methods.

	Year																					
Month	1995	1996	1997	1998	1999	2000	2001	2002	2003	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Chinook Salmon Harvest																						
May	141	45	--	--	--	--	509	522	--	--	--	474	--	--	--	--	--	--	--	--	--	--
June	907	2,165	--	--	--	--	6,630	4,028	--	--	--	7,708	4,797	--	--	--	--	--	7,071	7,071	5,229	1,273
July	505	1,097	--	254	--	1,611	3,260	2,822	--	--	--	3,080	3,873	--	--	--	--	--	4,016	4,016	3,800	2,964
Aug	245	257	--	35	--	108	299	474	--	1,207	--	688	503	--	--	--	--	--	1,669	1,669	1,078	1,593
Sept	15	62	--	6	--	93	61	59	--	203	--	20	4	--	--	--	--	--	--	--	--	--
TOTAL	1,813	3,626	--	295	--	1,812	10,759	7,905	--	1,410	--	11,970	9,177	--	--	--	--	--	12,756	12,757	10,108	5,997
Coho Salmon Harvest																						
May	0	0	--	--	--	--	0	0	--	--	--	0	--	--	--	--	--	--	--	--	--	--
June	335	474	--	--	--	--	1,771	1,332	--	--	--	3,913	3,686	--	--	--	--	--	4,436	4,436	1,839	126
July	402	3,565	--	1,937	--	568	4,736	9,798	--	--	--	15,982	17,043	--	--	--	--	--	15,101	15,101	22,871	5,758
Aug	1,907	3,771	--	1,500	--	2,125	6,488	8,292	--	15,647	--	13,791	16,459	--	--	--	--	--	12,612	12,612	14,004	4,580
Sept	1,115	1,417	--	79	--	790	1,006	1,010	--	3,215	--	733	3,136	--	--	--	--	--	--	--	--	--
TOTAL	3,759	9,227	--	3,516	--	3,483	14,001	20,432	--	18,862	--	34,419	40,324	--	--	--	--	--	32,148	32,149	38,713	10,464
Pacific Halibut Harvest																						
May	--	--	--	--	--	--	281	467	--	--	--	1,435	--	--	--	--	--	--	--	--	--	--
June	--	--	--	--	--	--	4,872	3,912	--	--	--	8,367	3,029	--	--	--	--	--	3,551	3,551	4,632	3,407
July	--	--	--	--	--	3,963	3,535	4,716	--	--	--	8,507	4,466	--	--	--	--	--	4,096	4,096	6,563	4,263
Aug	--	--	--	1,801	--	2,530	3,085	3,247	--	8,178	--	5,865	4,142	--	--	--	--	--	5,201	5,201	3,410	3,183
Sept	--	--	--	328	--	1,224	222	615	--	1,812	--	774	1,205	--	--	--	--	--	--	--	--	--
TOTAL	--	--	--	2,129	--	7,717	11,995	12,957	--	9,990	--	24,948	12,842	--	--	--	--	--	12,848	12,849	14,606	10,853

Over the same period, Chinook and Coho salmon harvests have increased by 253 and 1081 fish per year, on average (Table 28, Figure 15). Despite dramatic increases over the past 20 years, harvests in 2018 were the lowest that they have been since 1996. The Chinook Salmon harvest of 5,997 fish was less than 60% of that in 2016-17, and 47% of that in 2015. The Coho Salmon harvest of 10,464 fish was less than one third of that in 2015-17. Additionally, there were large declines in the numbers of the fish biosampled in 2018 as compared to other recent years (see Robichaud et al. 2016, 2017, Robichaud and Addison 2018). While the non-lodge harvest estimates are always potentially affected by unperceived sampling biases, the fact that harvest declines were observed for both non-lodge and lodge anglers suggests that estimated declines were real.

Conservation concerns for Chinook Salmon in 2018 resulted in a complete closure of the retention fishery from June 16 to July 9; and bag limits were reduced to one fish (half of the usual two fish limit) from June 1 to 15 and from July 10 to 31. Together, these management measures help explain the reduction in Chinook Salmon harvest in 2018. While the lower retention of Chinook Salmon might have resulted in elevated harvest of other species, there instead appeared to be a slight overall reduction in angling effort, with concomitant reduction in harvest of most other species. While there was more harvest of Pink Salmon in 2018 as compared to 2017, the most recent on-cycle year (2016) had more than 2.5 the harvest that was estimated for 2018. The only species whose harvest increased in 2018, compared to other recent years, is Lingcod. Harvest of Lingcod in 2018 was the highest it has been in the 2015-2018 period (3,051 in 2015 1,803 in 2016, 2,772 in 2017 and 4,270 in 2018). This, despite a coast-wide reduction in the Lingcod bag limit from three to two fish starting in May 2018.

Pacific Halibut harvest did not show a strong among-year trend (Figure 15). While fishing effort increased, the total harvest of Pacific Halibut in recent years (2009, 2015-2018) is very similar to that from 2001-2002; Table 28). This is very likely the result of effective management measures that were enacted between 2008 and 2009 (including the reduction in the daily bag limit from two to one), that appeared to have reduced the catch of halibut by nearly 50% (Figure 15).

The meta-analysis results make two important points: 1) the effort and CPUE of certain species has increased tremendously over the past ~20 years, potentially endangering stocks if left unchecked; and 2) management decisions from the recent past appear to have been effective, indicating that effective tools are available to keep effort and CPUE increases from going unchecked.

May Trailer Counts

Trailer counts made during May of 2018 revealed that substantial angling activity is being missed by starting the creel survey in June each year. In 2018, May effort was 72% of that in June, likely more than 2,000 boat trips. Much of the early season angling targets Chinook Salmon, and occurs during a period in which the Area 3 and 4 stock composition is unknown or poorly resolved. As such, the Department of Fisheries and Oceans Canada (DFO) has expressed interest in a May creel survey because genetic samples collected from Chinook Salmon would help them resolve stock composition, and help determine exploitation rates on early-run-timing Chinook Salmon stocks.

Precision of the Results

Overall, our level of survey effort, both for CPE and Effort estimation, were adequate, as evidenced by the relatively good precision of the harvest estimates for most of the taxa surveyed. In 2018 our precision goal was to estimate total harvests within 15% of the true value 19 times out of 20, for the main taxa (Chinook Salmon, Coho Salmon and Pacific Halibut), for which harvest estimates were precise to within 3-4%. Pink Salmon, Lingcod, groundfish, and rockfish estimates were precise to within 11%, 5%, 9%, and 5%, respectively. There were two species for which precision was worse than 15%: Chum and Sockeye salmon was precise to within 25% and 42%, respectively. The survey was not designed to meet any level of precision for these last species.

The precision of our Harvest estimates, depended largely on the variability in the CPE and effort data. Differing levels of angler expertise, and day-to-day changes in abundance of the target species, fishing effort, weather, or fishing conditions mean that fishing trips have a wide range of outcomes. Similarly, day-to-day differences in weather or other factors (such as leisure time, day of week, the price of gasoline, etc.) affect the number of anglers that decide to go out on the water on any given day, resulting in variability within the boat-count dataset. Sampling error is another main source of estimation imprecision; and by meeting our precision goals we have demonstrated that we are sampling an adequate proportion of the population.

Accuracy of the Results

The accuracy the results depends on the quality of data provided by the anglers to the interviewers. Each time an angler is relied upon for unconfirmed data, there is opportunity for results to become inaccurate. For example, in this study, 285 of the 733 Chinook Salmon (39%) and 346 of the 1,334 Coho Salmon (26%) reported as kept during angler interviews were recorded as observed by the interviewer. While not every fish can be observed², each one that goes unconfirmed is nevertheless a possible source of inaccuracy in the reported harvest totals.

In this study, the angler effort estimates depend on the accuracy of the aerial survey boat counts and the activity patterns for each month. For example, we have no estimate of the proportion of angling boats that are missed during overflights, where missed boats would contribute to the underestimation of recreational fishing effort. Also, we do not know if all fishing areas are adequately covered by the current flightpath – perhaps there are fishing locations that are hidden behind islands (e.g., Dunira Island or the Moffat Islands), too far from the flightpath for the surveyors to see active fishing boats. On the other hand, our estimate of fishing effort may be overestimated due to our inability to survey and count boats on foggy or bad weather days. We know that during bad weather, fewer anglers go out on the water; yet these same conditions can cause flights to be cancelled, and thus the lowest effort values are sometimes not sampled.

² Fish may not be observed when anglers refuse to provide access to their fish storage containers for various reasons. Reluctance to provide access may be due to fish already being processed and packed in ice, or boat launch traffic congestion causing tension, among other reasons.

Another potential source of inaccuracy is the removal of lodge boats from the overflight counts. There is uncertainty around whether all of the active lodge boats were observed by the overflights (e.g. vessels outside of the survey area). There is also uncertainty around the proportion of lodge vessels that were active during the overflights. These factors may influence the accuracy of the estimates of non-lodge fishing effort.

RECOMMENDATIONS

- The catch and effort estimates for the Area 3 and 4 recreational creel surveys for 2010-2014 should be compiled and included in summary tables similar to Table 27 and 28 in this report. If sufficient funding can be obtained, the results from 2010-14 surveys should be compiled into a manuscript report. This additional information will help provide context for the current results, and help clarify how rates of effort and harvest have changed in recent years.
- We recommend a four-month survey window, May to August (rather than three months from June to August). May surveys will be important because there is a significant amount of heretofore unquantified angling effort and harvest during that month (trailer counts in May 2018 suggested that there may have been 2000 boat trips; Figure 9). The Department of Fisheries and Oceans Canada (DFO) has expressed interest in a May creel survey because genetic samples collected from Chinook Salmon will help resolve the currently unknown early-season Area 3 and 4 stock composition, which will help determine exploitation rates on early-run-timing Chinook Salmon stocks.

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We would like to thank the creel surveyors (Katelyn Cooper, Jesse Beaubier-Brulotte, Kolby Jones, Steven Grainger, and Sarah Jordan) for collecting large amounts of data and biosamples from anglers. Thanks to Pacific Salmon Commission, who provided support for this work through the Northern Fund. Thanks to Darren Chow and Peter Katinic for acting as liaison between our team and DFO. Thanks also go to the administrative staff at LGL and NCSFNSS, whose work made it possible for this study to be conducted. We also would like to thank Karl English for providing useful comments and edits on previous drafts of this report. Last but not least, we thank the many anglers, charter operators, and fishing lodges who participated in the survey.

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APPENDICES

APPENDIX 1: DATA FORMS.

2018 AREA 3/4 CREEL SURVEY
AERIAL SURVEY VESSEL ENUMERATION FORM

DATE: _____ FLIGHT#: _____

SURVEYOR: _____

Note: only survey shaded subareas when conditions permit

Sub Area	RIGHT FRONT COUNT				TOTAL COUNT				FLIGHT TIME		SEA STATE (1-4)
	SPORT (Fishing)	SPORT (Moving)	COMM. (Fishing)	COMM. (Moving)	S (F)	S (M)	C (F)	C (M)	START	FINISH	
A											
C											
B											
C											
D											
E											
C											
F											
E											
F											
G											
L											
G											
J											
I											
K											
I											
H											
C											
A											

WEATHER / SEA STATE / COMMENTS:

SIGNATURE: _____

SHIFT SUMMARY SHEET

Landing Site: _____ **Interviewer Name:** _____

Interviewer Code: _____ **Date:** _____

Shift Code: _____ **Shift Times:** _____

Time Blocks	
From - To	TB
Before 7:00	1
7:00 - 7:59	2
8:00 - 8:59	3
9:00 - 9:59	4
10:00 - 10:59	5
11:00 - 11:59	6
12:00 - 12:59	7
13:00 - 13:59	8
14:00 - 14:59	9
15:00 - 15:59	10
16:00 - 16:59	11
17:00 - 17:59	12
18:00 - 18:59	13
19:00 - 19:59	14
20:00 - 20:59	15
21:00 - 21:59	16

* Fill in using numbers only.

Time Block	Fishing Conditions	Boats Missed	Not Fishing	Fishing

Totals:

--	--	--

Comments: _____

Trailer Count: _____ **Time:** _____

Database Tally # :

(office use only)

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APPENDIX 2: HPE ESTIMATES (AVERAGE NUMBER OF FISH HARVESTED PER BOAT TRIP) BY MONTH, TRIP TYPE, SUBAREA AND SPECIES.

Average number of salmon harvested per boat trip in June 1-15, 2018, by trip type and subarea.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	2.27	1.83	2.80	0.06	0.37	1.23	1.06	1.06	0.92	0.83	0.91	0.83
Unguided	1.83	1.83	1.83	0.04	0.50	0.29	0.41	0.66	0.72	0.55	0.67	0.55
Combined	2.12	2.13	2.33	0.05	0.47	0.54	0.62	0.76	0.77	0.71	0.77	0.20
Chum Salmon												
Guided												
Unguided												
Combined												
Coho Salmon												
Guided												
Unguided								0.02	0.03			
Combined								0.02	0.02			
Pink Salmon												
Guided	0.09		0.20		0.02	0.08						
Unguided												
Combined	0.06		0.17			0.02						
Sockeye Salmon												
Guided												
Unguided												
Combined												
Unknown Salmon												
Guided												
Unguided												
Combined												

Average number of salmon harvested per boat trip in June 16 - July 9, 2018, by trip type and subarea.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided												
Unguided						0.04			0.09	0.08		0.02
Combined						0.03			0.04	0.06		
Chum Salmon												
Guided												
Unguided						0.02						
Combined						0.02						
Coho Salmon												
Guided	0.33		0.27				0.10	0.10	0.11		0.03	0.19
Unguided						0.08		0.43	0.24		0.03	0.02
Combined	0.14		0.23			0.06		0.30	0.17		0.03	0.17
Pink Salmon												
Guided	0.11		0.05									
Unguided		0.08							0.06			
Combined	0.05	0.04							0.03			
Sockeye Salmon												
Guided												
Unguided												
Combined												
Unknown Salmon												
Guided												
Unguided												
Combined												

Average number of salmon harvested per boat trip in July 10-31, 2018, by trip type and subarea.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	1.33	1.22	1.33		0.20	1.00	1.63	1.63	1.63	2.13	1.06	1.25
Unguided	0.63	1.55	1.00		0.17	0.41	0.31	0.50	0.75	0.80	0.36	1.00
Combined	0.67	1.40	1.40		0.17	0.48	0.31	0.50	1.05	1.62	0.62	1.19
Chum Salmon												
Guided							0.02	0.02	0.02			
Unguided	0.25								0.01			
Combined	0.17								0.01			
Coho Salmon												
Guided	2.73	3.33	2.73		0.07	0.33	3.92	3.92	3.92	4.38	1.82	8.45
Unguided	1.00	1.73	1.38		0.11	0.27	0.17	2.20	1.44	0.60	0.50	2.83
Combined	1.08	2.45	1.70		0.11	0.27	0.17	2.20	2.29	2.92	1.00	7.15
Pink Salmon												
Guided	1.27	1.78	1.27		0.04	0.22	0.35	0.35	0.35		0.06	0.15
Unguided	0.38		0.13		0.06	0.13	0.03	0.10	0.30		0.54	0.17
Combined	0.33	0.80	0.30		0.06	0.14	0.03	0.10	0.32		0.36	0.15
Sockeye Salmon												
Guided												
Unguided												
Combined												
Unknown Salmon												
Guided												
Unguided												
Combined												

Average number of salmon harvested per boat trip in August 2018, by trip type and subarea.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	0.92	0.69	0.62		0.02	0.08	0.96	0.96	1.00	0.41	0.75	0.27
Unguided	0.80	0.43	0.50		0.14	0.17	0.21	0.36	0.47	0.31	0.42	
Combined	0.85	0.57	0.54		0.13	0.16	0.24	0.54	0.69	0.37	0.53	0.23
Chum Salmon												
Guided		0.08	0.08				0.02	0.02	0.02	0.08	0.17	0.04
Unguided	0.05		0.04				0.04	0.02		0.03		
Combined	0.03	0.04	0.05				0.04	0.02	0.01	0.06	0.06	0.03
Coho Salmon												
Guided	2.77	2.23	3.38		0.15	0.67	1.34	1.34	1.37	1.56	2.08	1.38
Unguided	1.50	1.33	2.08		0.18	0.24	1.11	1.06	1.05	0.25	0.33	
Combined	2.00	1.83	2.51		0.18	0.30	1.10	1.15	1.18	0.97	0.92	1.16
Pink Salmon												
Guided		0.12	1.08				0.24	0.24	0.26	0.18		0.27
Unguided		0.19	0.19		0.02	0.06	0.02	0.08	0.13	0.03		
Combined		0.15	0.49		0.02	0.05	0.02	0.13	0.18	0.11		0.23
Sockeye Salmon												
Guided		0.04	0.08									
Unguided			0.04									
Combined		0.02	0.05									
Unknown Salmon												
Guided												
Unguided												
Combined												

Average number of groundfish harvested per boat trip in June 1-15, 2018, by trip type and subarea.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow Flounder												
Guided										0.06		0.06
Unguided												
Combined										0.14		
Big Skate												
Guided												
Unguided												
Combined												
Cabezon												
Guided							0.13	0.13	0.17	0.06		0.06
Unguided												
Combined								0.03	0.05			0.20
English Sole												
Guided												
Unguided												
Combined												
Greenling												
Guided							0.13	0.13	0.17	0.11		0.11
Unguided										0.05	0.07	0.05
Combined								0.03	0.05	0.29	0.04	
Lingcod												
Guided	1.27	1.00	1.60		0.06	0.23	2.00	2.00	1.75	0.83	1.00	0.83
Unguided								0.14	0.22	0.20	0.27	0.20
Combined	0.82	0.75	1.33			0.06	0.52	0.59	0.64	0.29	0.58	0.40
Pacific Cod												
Guided	0.27		0.60				0.56	0.56	0.75	0.44	0.73	0.44
Unguided						0.14	0.06	0.02		0.10	0.13	0.10
Combined	0.18		0.50			0.10	0.05	0.15	0.20		0.38	
Pacific Halibut												
Guided	1.82	3.17	0.20		0.16	0.62	1.50	1.50	1.83	1.78	1.91	1.78
Unguided	0.67	0.67	0.67		0.06	0.54	0.29	0.58	0.72	1.10	0.87	1.10
Combined	1.41	2.38	0.17		0.05	0.56	0.33	0.80	1.02	1.00	1.31	2.60
Ratfish												
Guided												
Unguided												
Combined												
Rock Sole												
Guided												
Unguided												
Combined												
Pacific Halibut												
Guided	1.82	3.17	0.20		0.16	0.62	1.50	1.50	1.83	1.78	1.91	1.78
Unguided	0.67	0.67	0.67		0.06	0.54	0.29	0.58	0.72	1.10	0.87	1.10
Combined	1.41	2.38	0.17		0.05	0.56	0.33	0.80	1.02	1.00	1.31	2.60
Sablefish												
Guided							0.06	0.06	0.08	0.17	0.27	0.17
Unguided												
Combined								0.02	0.02		0.12	
Starry Flounder												
Guided												
Unguided												
Combined												
Unknown Groundfish												
Guided					0.02	0.08						
Unguided					0.11							
Combined					0.11	0.02						

Average number of groundfish harvested per boat trip in June 16 - July 9, 2018, by trip type and subarea.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow Flounder												
Guided	0.11		0.05									
Unguided												
Combined	0.05											
Big Skate												
Guided												
Unguided												
Combined												
Cabezon												
Guided												
Unguided												
Combined												
English Sole												
Guided												
Unguided											0.03	0.02
Combined											0.01	
Greenling												
Guided												
Unguided		0.23							0.03			
Combined		0.13							0.01			
Lingcod												
Guided		3.10	1.45		0.01	0.08	1.06	1.06	0.95		0.80	1.76
Unguided	0.08	0.85	0.10		0.05	0.04	0.10	0.14	0.97	1.08	0.32	0.56
Combined	0.05	1.83	0.15		0.05	0.05	0.21	1.10	0.96	0.72	0.58	1.74
Pacific Cod												
Guided	0.33	0.20	0.23		0.05	0.38	0.12	0.12	0.11		0.13	0.05
Unguided	0.08								0.24		0.06	0.04
Combined	0.19	0.09				0.08	0.07		0.17		0.09	0.04
Pacific Halibut												
Guided	2.56	2.60	2.59		0.25	2.00	2.51	2.51	2.57	3.33	2.30	3.19
Unguided	1.58	1.92	1.20		0.34	0.82	0.30	2.29	1.09	1.17	1.21	1.21
Combined	2.00	2.22	1.54		0.34	1.06	0.93	2.10	1.92	1.89	1.80	3.04
Ratfish												
Guided												
Unguided												
Combined												
Rock Sole												
Guided							0.02	0.02	0.02			
Unguided												
Combined									0.01			
Pacific Halibut												
Guided	2.56	2.60	2.59		0.25	2.00	2.51	2.51	2.57	3.33	2.30	3.19
Unguided	1.58	1.92	1.20		0.34	0.82	0.30	2.29	1.09	1.17	1.21	1.21
Combined	2.00	2.22	1.54		0.34	1.06	0.93	2.10	1.92	1.89	1.80	3.04
Sablefish												
Guided	0.22		0.09		0.02	0.15	0.12	0.12	0.14		0.35	
Unguided			0.10			0.12			0.06			
Combined	0.10		0.08			0.13			0.10		0.19	
Starry Flounder												
Guided												
Unguided												
Combined												
Unknown Groundfish												
Guided												
Unguided												
Combined												

Average number of groundfish harvested per boat trip in July 10-31, 2018, by trip type and subarea.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow Flounder												
Guided												
Unguided												
Combined												
Big Skate												
Guided												
Unguided												
Combined												
Cabezon												
Guided												
Unguided									0.02			
Combined									0.01			
English Sole												
Guided												
Unguided					0.02							
Combined					0.02							
Greenling												
Guided											0.47	
Unguided		0.18							0.02		0.04	
Combined		0.10							0.01		0.20	
Lingcod												
Guided	1.00	1.67	1.00		0.09	0.44	0.27	0.27	0.27			0.30
Unguided		0.55				0.07	0.07	0.80	0.39		0.18	0.50
Combined		1.05				0.11	0.07	0.80	0.35		0.11	0.35
Pacific Cod												
Guided					0.02					0.13	0.12	0.05
Unguided					0.02	0.03			0.05		0.04	
Combined					0.03	0.02			0.03	0.08	0.07	0.04
Pacific Halibut												
Guided	1.40	1.11	1.40		0.22	1.00	1.50	1.50	1.50	0.88	1.76	3.40
Unguided	1.13	0.73			0.13	0.53	0.86	1.10	0.41		1.00	1.67
Combined	1.17	0.90	0.60		0.14	0.58	0.86	1.10	0.78	0.54	1.29	3.00
Ratfish												
Guided												
Unguided												
Combined												
Rock Sole												
Guided												
Unguided					0.02							
Combined					0.02							
Pacific Halibut												
Guided	1.40	1.11	1.40		0.22	1.00	1.50	1.50	1.50	0.88	1.76	3.40
Unguided	1.13	0.73			0.13	0.53	0.86	1.10	0.41		1.00	1.67
Combined	1.17	0.90	0.60		0.14	0.58	0.86	1.10	0.78	0.54	1.29	3.00
Sablefish												
Guided							0.17	0.17	0.17			
Unguided							0.07		0.25			
Combined							0.07		0.22			
Starry Flounder												
Guided												
Unguided												
Combined												
Unknown Groundfish												
Guided												
Unguided				0.39		0.01						
Combined				0.21		0.01						

Average number of groundfish harvested per boat trip in August 2018, by trip type and subarea.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow Flounder												
Guided												
Unguided												
Combined												
Big Skate												
Guided												
Unguided								0.01	0.02			
Combined								0.01	0.01			
Cabezon												
Guided												
Unguided												
Combined												
English Sole												
Guided												
Unguided								0.02	0.03			
Combined								0.01	0.02			
Greenling												
Guided	0.08						0.04	0.04	0.04			
Unguided							0.04	0.02				
Combined	0.03						0.04	0.02	0.02			
Lingcod												
Guided	0.08	1.12	0.08				0.46	0.46	0.50	0.64	0.58	0.69
Unguided		0.52		0.07		0.03	0.11	0.21	0.28	0.22	0.13	0.80
Combined	0.03	0.85	0.03	0.05		0.02	0.10	0.28	0.37	0.45	0.28	0.71
Pacific Cod												
Guided							0.02	0.02	0.02	0.03	0.08	
Unguided		0.10			0.05	0.03	0.15	0.06				
Combined		0.04			0.04	0.02	0.14	0.05	0.01	0.01	0.03	
Pacific Halibut												
Guided	0.62	1.88	0.85		0.27	1.25	2.18	2.18	2.22	2.13	1.58	2.46
Unguided	0.80	1.24	0.46		0.39	0.83	0.64	0.70	0.75	0.94	0.79	2.20
Combined	0.73	1.60	0.59		0.38	0.89	0.73	1.15	1.36	1.59	1.06	2.42
Ratfish												
Guided												
Unguided												
Combined												
Rock Sole												
Guided												
Unguided												
Combined												
Pacific Halibut												
Guided	0.62	1.88	0.85		0.27	1.25	2.18	2.18	2.22	2.13	1.58	2.46
Unguided	0.80	1.24	0.46		0.39	0.83	0.64	0.70	0.75	0.94	0.79	2.20
Combined	0.73	1.60	0.59		0.38	0.89	0.73	1.15	1.36	1.59	1.06	2.42
Sablefish												
Guided												
Unguided	0.10				0.02	0.04	0.02	0.09	0.14			
Combined	0.06				0.02	0.04	0.02	0.06	0.08			
Starry Flounder												
Guided												
Unguided												
Combined												
Unknown Groundfish												
Guided												
Unguided				0.14	0.05							
Combined				0.09	0.04							

Average number of rockfish harvested per boat trip in June 1-15, 2018, by trip type and subarea.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided					0.02	0.08	0.63	0.63	0.58	0.33		0.33
Unguided						0.06						
Combined						0.06	0.14	0.15	0.16	0.86		
Boccacio Rockfish												
Guided												
Unguided												
Combined												
Canary Rockfish												
Guided												
Unguided												
Combined												
China Rockfish												
Guided										0.22	0.36	0.22
Unguided												
Combined											0.15	
Copper Rockfish												
Guided							0.13	0.13	0.17			
Unguided						0.34						
Combined						0.25		0.03	0.05			
Dusky Rockfish												
Guided												
Unguided												
Combined												
Quillback Rockfish												
Guided							0.06	0.06	0.08	0.28	0.09	0.28
Unguided				0.08		0.09		0.14	0.22	0.15	0.20	0.15
Combined				0.03		0.06		0.12	0.18	0.57	0.15	
Redband Rockfish												
Guided												
Unguided												
Combined												
Redstripe Rockfish												
Guided												
Unguided												
Combined												
Tiger Rockfish												
Guided							0.06	0.06	0.08			
Unguided												
Combined								0.02	0.02			
Widow Rockfish												
Guided												
Unguided												
Combined												
Yelloweye Rockfish												
Guided	0.09	0.17										
Unguided	0.17	0.17	0.17					0.02	0.03			
Combined	0.12	0.13						0.02	0.02			
Yellowtail Rockfish												
Guided					0.02	0.08						
Unguided												
Combined						0.02						
Unknown rockfish												
Guided							1.38	1.38	1.42	2.06	2.09	2.06
Unguided	0.50	0.50	0.50				1.12	0.86	0.75	0.15	0.20	0.15
Combined	0.18						1.14	0.98	0.93	0.86	1.00	1.60

Average number of rockfish harvested per boat trip in June 16 – July 9, 2018, by trip type and subarea.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided		0.20	0.09				0.02	0.02			0.33	
Unguided	0.17								0.15	0.08		0.02
Combined	0.10	0.09					0.07		0.06	0.06	0.18	
Boccacio Rockfish												
Guided												
Unguided												
Combined												
Canary Rockfish												
Guided												
Unguided												
Combined												
China Rockfish												
Guided							0.06	0.06	0.07	0.33		
Unguided												
Combined									0.04	0.11		
Copper Rockfish												
Guided					0.01	0.08	0.08	0.08			0.10	
Unguided												
Combined						0.02	0.29				0.05	
Dusky Rockfish												
Guided												
Unguided								0.86				
Combined								0.60				
Quillback Rockfish												
Guided		0.60	0.27		0.05	0.38	0.65	0.65	0.75		0.50	0.10
Unguided	0.25	0.38	0.50	0.10	0.12	0.02	0.30		0.24		0.06	0.04
Combined	0.14	0.48	0.38	0.04	0.12	0.09	0.21		0.53		0.30	0.09
Redband Rockfish												
Guided												
Unguided												
Combined												
Redstripe Rockfish												
Guided												
Unguided												
Combined												
Tiger Rockfish												
Guided							0.06	0.06	0.02			
Unguided												
Combined							0.14		0.01			
Widow Rockfish												
Guided												
Unguided												
Combined												
Yelloweye Rockfish												
Guided												
Unguided		0.08			0.05							
Combined		0.04			0.05							
Yellowtail Rockfish												
Guided												
Unguided					0.02							
Combined					0.02							
Unknown rockfish												
Guided	0.11	0.80	1.32		0.11	0.85	1.22	1.22	1.25	1.17	0.98	2.81
Unguided	1.17	1.31			0.29	0.51	0.50	0.71	0.59	0.50	0.59	0.58
Combined	0.71	1.09	1.54		0.29	0.58	0.50	1.00	0.96	0.72	0.80	2.65

Average number of rockfish harvested per boat trip in July 10-31, 2018, by trip type and subarea.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided	0.27	0.44	0.27				0.02	0.02	0.02			
Unguided					0.05	0.04			0.04			
Combined		0.20			0.05	0.04			0.03			
Boccacio Rockfish												
Guided												
Unguided												
Combined												
Canary Rockfish												
Guided												
Unguided												
Combined												
China Rockfish												
Guided												
Unguided						0.04						
Combined						0.04						
Copper Rockfish												
Guided												
Unguided						0.03			0.01			
Combined						0.02			0.01			
Dusky Rockfish												
Guided												
Unguided					0.02							
Combined					0.02							
Quillback Rockfish												
Guided							0.12	0.12	0.12	0.50	0.29	0.05
Unguided		0.09			0.16	0.16			0.05		0.39	
Combined		0.05			0.16	0.14			0.07	0.31	0.36	0.04
Redband Rockfish												
Guided												
Unguided									0.01			
Combined									0.01			
Redstripe Rockfish												
Guided												
Unguided									0.02			
Combined									0.01			
Tiger Rockfish												
Guided												
Unguided												
Combined												
Widow Rockfish												
Guided												
Unguided												
Combined												
Yelloweye Rockfish												
Guided												
Unguided						0.01						
Combined						0.01						
Yellowtail Rockfish												
Guided												
Unguided					0.05	0.07						
Combined					0.05	0.06						
Unknown rockfish												
Guided	1.60	2.44	1.60		0.22	1.11	0.40	0.40	0.40			0.35
Unguided		1.64	0.25		0.25	0.13	0.69	1.30	0.37		0.46	
Combined		2.00	0.40		0.25	0.24	0.69	1.30	0.38		0.29	0.27

Average number of rockfish harvested per boat trip in August 2018, by trip type and subarea.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided	0.08	0.42	0.15				0.22	0.22	0.24	0.10		0.15
Unguided			0.15			0.03	0.13	0.07	0.03	0.44		2.40
Combined	0.03	0.23	0.15			0.02	0.12	0.12	0.12	0.25		0.52
Boccacio Rockfish												
Guided												
Unguided												
Combined												
Canary Rockfish												
Guided							0.02	0.02	0.02			
Unguided												
Combined								0.01	0.01			
China Rockfish												
Guided		0.04					0.02	0.02	0.02			
Unguided	0.05	0.05					0.04	0.04	0.03			
Combined	0.03	0.04					0.04	0.03	0.03			
Copper Rockfish												
Guided		0.04					0.04	0.04	0.04	0.03		0.04
Unguided							0.06	0.03				
Combined		0.02					0.06	0.03	0.02	0.01		0.03
Dusky Rockfish												
Guided												
Unguided							0.02	0.01				
Combined							0.02	0.01				
Quillback Rockfish												
Guided		0.23			0.02	0.08	0.34	0.34	0.37	0.10	0.17	0.08
Unguided		0.29		0.01	0.02	0.21	0.21	0.18	0.16	0.22	0.17	
Combined		0.26		0.01	0.02	0.20	0.20	0.23	0.25	0.15	0.17	0.06
Redband Rockfish												
Guided												
Unguided												
Combined												
Redstripe Rockfish												
Guided												
Unguided												
Combined												
Tiger Rockfish												
Guided		0.08								0.03		0.04
Unguided		0.05					0.02	0.02	0.02			
Combined		0.06					0.02	0.01	0.01	0.01		0.03
Widow Rockfish												
Guided												
Unguided												
Combined												
Yelloweye Rockfish												
Guided												
Unguided								0.01	0.02			
Combined								0.01	0.01			
Yellowtail Rockfish												
Guided												
Unguided						0.04	0.02	0.04	0.06			
Combined						0.04	0.02	0.03	0.04			
Unknown rockfish												
Guided	0.23	0.50	0.08				0.60	0.60	0.63	1.08		1.62
Unguided		0.48	0.35	0.06		0.11	0.28	0.24	0.22	0.16	0.17	0.20
Combined	0.09	0.49	0.26	0.04		0.10	0.27	0.35	0.39	0.66	0.11	1.39

APPENDIX 3: COUNTS OF ACTIVELY FISHING SPORT VESSELS IN EACH SUBAREA, BY DATE

Counts of actively fishing sport vessels in each subarea, by date, 2018, along with numbers that were estimated to be lodge boats. Weekend and holiday flights are shaded.

Month	Date	Subarea												Area 3
		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Lodge Boats
June	3 Jun 2018	3	12	3	1	6	12	1	0	4	1	5	1	12
	6 Jun 2018	3	2	5	0	4	2	1	0	0	0	0	0	11
	10 Jun 2018	4	17	9	2	6	9	8	1	5	0	4	7	12
	15 Jun 2018	13	7	3	0	2	7	7	1	2	3	5	0	4
	18 Jun 2018	1	5	0	1	1	8	0	0	1	6	8	2	0
	23 Jun 2018	3	16	4	0	9	15	1	0	11	6	8	4	12
	28 Jun 2018	4	6	6	4	0	2	2	1	6	0	6	0	12
	30 Jun 2018	25	11	6	1	2	7	0	3	10	4	4	3	6
July	4 Jul 2018	3	15	2	3	0	22	6	4	16	8	3	8	8
	10 Jul 2018	6	13	2	3	9	21	0	7	13	5	6	7	12
	13 Jul 2018	18	6	8	1	3	33	3	2	30	1	11	0	4
	17 Jul 2018	7	18	13	3	1	17	1	1	23	5	2	5	8
	21 Jul 2018	19	15	13	4	2	15	0	0	16	0	6	0	8
	29 Jul 2018	15	12	12	9	2	19	1	0	40	0	7	0	12
August	1 Aug 2018	9	12	13	5	0	8	0	0	16	0	3	0	12
	5 Aug 2018	19	13	6	2	3	33	2	9	36	8	5	10	4
	8 Aug 2018	6	16	13	3	6	15	2	2	6	7	2	8	4
	10 Aug 2018	7	2	11	3	4	12	1	0	18	0	3	0	0
	16 Aug 2018	8	12	8	3	4	10	3	1	10	6	5	0	0
	18 Aug 2018	3	13	14	1	4	14	1	2	4	0	1	8	0
	23 Aug 2018	13	6	11	2	3	7	2	0	5	1	2	1	4
	25 Aug 2018	12	3	20	2	1	8	0	0	2	1	3	0	4
	30 Aug 2018	8	4	9	2	0	7	0	1	2	2	1	0	3

APPENDIX 4: RETAINED AND RELEASED FISH (LODGE REPORTS AND NON-LODGE CREEL ESTIMATES) BY MONTH, SUBAREA AND SPECIES.

Estimated numbers of Chinook Salmon retained by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	246.1	231.9	108.3	0.6	50.1	102.1	69.1	10.1	50.0	29.9	65.1	6.1	969
Lodge Report	127.6	120.3	56.2	0	0	0	0	0	0	0	0	0	304
JUNE RETAINED	373.7	352.1	164.4	0.6	50.1	102.1	69.1	10.1	50.0	29.9	65.1	6.1	1,273
SE	11.5	42.8	26.5	0.4	12.1	19.3	16.9	0.4	8.9	13.1	12.9	6.1	63
July													
Creel Estimate	214.5	413.0	308.7	0	21.9	353.0	11.7	41.9	837.3	153.1	129.4	119.9	2,604
Lodge Report	82.5	158.8	118.7	0	0	0	0	0	0	0	0	0	360
JULY RETAINED	297.0	571.9	427.4	0	21.9	353.0	11.7	41.9	837.3	153.1	129.4	119.9	2,964
SE	53.3	74.1	69.5	0	6.4	62.5	3.8	22.5	61.8	41.3	23.9	33.5	158
August													
Creel Estimate	320.0	208.8	254.6	0	17.5	91.4	13.3	39.5	343.4	46.5	66.9	29.8	1,432
Lodge Report	65.8	42.9	52.3	0	0	0	0	0	0	0	0	0	161
AUGUST RETAINED	385.7	251.7	306.9	0	17.5	91.4	13.3	39.5	343.4	46.5	66.9	29.8	1,593
SE	59.7	43.0	48.7	0	6.1	26.1	4.3	0.6	38.0	1.0	17.3	11.6	102
TOTAL													
Creel Estimate	781	854	672	1	89	546	94	91	1,231	230	261	156	5,005
Lodge Report	276	322	227	0	0	0	0	0	0	0	0	0	825
TOTAL RETAINED	1,056	1,176	899	1	89	546	94	91	1,231	230	261	156	5,830
SE	81	96	89	0	15	70	18	23	73	43	32	36	198

Estimated numbers of Chinook Salmon released by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	422.9	240.9	128.4	0.3	131.1	92.8	68.1	36.0	251.2	6.9	104.4	44.4	1,528
Lodge Report	316.5	180.3	96.1	0	0	0	0	0	0	0	0	0	593
JUNE RELEASED	739.5	421.3	224.6	0.3	131.1	92.8	68.1	36.0	251.2	6.9	104.4	44.4	2,121
SE	80.4	73.0	38.2	0.3	54.7	23.8	24.1	15.8	34.8	5.0	22.6	23.6	141
July													
Creel Estimate	223.8	238.5	240.3	0.2	12.8	480.3	27.6	20.1	593.8	131.8	146.8	89.2	2,205
Lodge Report	129.7	138.1	139.2	0	0	0	0	0	0	0	0	0	407
JULY RELEASED	353.5	376.6	379.5	0.2	12.8	480.3	27.6	20.1	593.8	131.8	146.8	89.2	2,612
SE	79.0	72.8	64.8	0.2	4.9	117.2	11.3	9.1	70.9	51.8	31.3	31.0	198
August													
Creel Estimate	320.0	116.0	448.6	0	8.7	84.4	3.3	15.7	144.6	5.4	0	12.8	1,159
Lodge Report	38.0	13.8	53.3	0	0	0	0	0	0	0	0	0	105
AUGUST RELEASED	357.9	129.8	501.9	0	8.7	84.4	3.3	15.7	144.6	5.4	0	12.8	1,264
SE	111.1	26.0	92.8	0	6.7	25.7	2.6	0.2	29.1	0.1	0	6.9	152
TOTAL													
Creel Estimate	967	595	817	1	153	657	99	72	990	144	251	146	4,892
Lodge Report	484	332	289	0	0	0	0	0	0	0	0	0	1,105
TOTAL RELEASED	1,451	928	1,106	1	153	657	99	72	990	144	251	146	5,997
SE	158	106	119	0	55	122	27	18	84	52	39	40	287

Estimated numbers of Coho Salmon retained by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	17.9	0	10.5	0	0	17.6	0	12.7	38.8	0	4.1	15.3	117
Lodge Report	5.7	0	3.3	0	0	0	0	0	0	0	0	0	9
JUNE RETAINED	23.5	0	13.8	0	0	17.6	0	12.7	38.8	0	4.1	15.3	126
SE	11.2	0	8.0	0	0	10.1	0	6.6	9.5	0	2.1	8.7	22
July													
Creel Estimate	362.1	722.8	382.3	0	13.9	210.6	6.5	191.8	1833.1	269.4	210.4	728.6	4,932
Lodge Report	203.8	406.9	215.2	0	0	0	0	0	0	0	0	0	826
JULY RETAINED	565.9	1129.8	597.5	0	13.9	210.6	6.5	191.8	1833.1	269.4	210.4	728.6	5,758
SE	74.7	157.7	137.5	0	5.0	48.5	3.8	131.5	167.3	111.2	50.6	115.8	354
August													
Creel Estimate	754.2	665.1	1188.2	0	23.3	175.8	61.9	83.5	587.4	123.4	116.2	153.3	3,932
Lodge Report	187.4	165.3	295.3	0	0	0	0	0	0	0	0	0	648
AUGUST RETAINED	941.6	830.4	1483.5	0	23.3	175.8	61.9	83.5	587.4	123.4	116.2	153.3	4,580
SE	136.8	110.9	159.3	0	6.6	34.6	13.5	1.2	43.1	2.8	31.8	45.2	251
TOTAL													
Creel Estimate	1,134	1,388	1,581	0	37	404	68	288	2,459	393	331	897	8,981
Lodge Report	397	572	514	0	0	0	0	0	0	0	0	0	1,483
TOTAL RETAINED	1,531	1,960	2,095	0	37	404	68	288	2,459	393	331	897	10,464
SE	156	193	211	0	8	60	14	132	173	111	60	125	434

Estimated numbers of Coho Salmon released by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	0	0	0	0	0	7.0	0	2.9	0	2.0	0	12
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	0	0	0	0	0	7.0	0	2.9	0	2.0	0	12
SE	0	0	0	0	0	0	6.3	0	2.1	0	1.5	0	7
July													
Creel Estimate	160.9	88.5	0	0	2.0	8.7	5.0	0	111.0	0	1.2	23.2	401
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	160.9	88.5	0	0	2.0	8.7	5.0	0	111.0	0	1.2	23.2	401
SE	118.5	67.7	0	0	1.7	7.2	3.4	0	37.3	0	0.9	16.1	143
August													
Creel Estimate	11.4	0	84.9	0	0	0	16.6	9.9	31.6	0	0	0	154
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	11.4	0	84.9	0	0	0	16.6	9.9	31.6	0	0	0	154
SE	8.3	0	30.6	0	0	0	10.8	0.1	12.5	0	0	0	36
TOTAL													
Creel Estimate	172	89	85	0	2	9	29	10	146	0	3	23	567
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	172	89	85	0	2	9	29	10	146	0	3	23	567
SE	119	68	31	0	2	7	13	0	39	0	2	16	147

Estimated numbers of Pink Salmon retained by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	12.8	7.4	7.7	0	0	3.6	0	0	5.8	0	0	0	37
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	12.8	7.4	7.7	0	0	3.6	0	0	5.8	0	0	0	37
SE	5.1	5.2	5.5	0	0	2.5	0	0	2.9	0	0	0	10
July													
Creel Estimate	111.8	240.8	66.1	0	7.9	104.3	1.3	8.4	253.2	0	73.9	15.5	883
Lodge Report	3.5	7.5	2.1	0	0	0	0	0	0	0	0	0	13
JULY RETAINED	115.2	248.3	68.2	0	7.9	104.3	1.3	8.4	253.2	0	73.9	15.5	896
SE	60.8	87.7	34.7	0	3.4	34.0	1.2	8.4	37.5	0	22.7	7.3	126
August													
Creel Estimate	0	54.1	230.4	0	2.9	28.1	1.1	9.4	90.4	14.3	0	29.8	461
Lodge Report	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	0	54.1	230.4	0	2.9	28.1	1.1	9.4	90.4	14.3	0	29.8	461
SE	0	18.2	91.0	0	2.2	12.1	0.9	0.1	16.2	1.2	0	10.7	96
TOTAL													
Creel Estimate	125	302	304	0	11	136	2	18	349	14	74	45	1,381
Lodge Report	3	7	2	0	0	0	0	0	0	0	0	0	13
TOTAL RETAINED	128	310	306	0	11	136	2	18	349	14	74	45	1,394
SE	61	90	98	0	4	36	1	8	41	1	23	13	158

Estimated numbers of Pink Salmon released by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	0	0	0	0	3.6	0	0	0	0	0	0	4
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	0	0	0	0	3.6	0	0	0	0	0	0	4
SE	0	0	0	0	0	2.5	0	0	0	0	0	0	3
July													
Creel Estimate	107.3	29.5	88.2	0	2.0	52.1	0	0	104.0	85.1	23.1	0	491
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	107.3	29.5	88.2	0	2.0	52.1	0	0	104.0	85.1	23.1	0	491
SE	79.0	22.6	36.0	0	1.7	22.3	0	0	29.6	71.1	12.9	0	121
August													
Creel Estimate	0	30.9	24.2	0	5.8	28.1	0	0.9	9.0	0	0	0	99
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	30.9	24.2	0	5.8	28.1	0	0.9	9.0	0	0	0	99
SE	0	13.7	12.3	0	4.5	15.7	0	0.0	4.6	0	0	0	25
TOTAL													
Creel Estimate	107	60	112	0	8	84	0	1	113	85	23	0	594
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	107	60	112	0	8	84	0	1	113	85	23	0	594
SE	79	26	38	0	5	27	0	0	30	71	13	0	124

Estimated numbers of Chum Salmon retained by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	0	0	0	0	4.4	0	0	0	0	0	0	4
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	0	0	0	0	0	4.4	0	0	0	0	0	0	4
SE	0	0	0	0	0	3.2	0	0	0	0	0	0	3
July													
Creel Estimate	53.6	0	0	0	0	2.7	0	0	10.4	0	0	0	67
Lodge Report	14.3	0	0	0	0	0	0	0	0	0	0	0	14
JULY RETAINED	68.0	0	0	0	0	2.7	0	0	10.4	0	0	0	81
SE	39.5	0	0	0	0	1.9	0	0	5.6	0	0	0	40
August													
Creel Estimate	11.4	15.5	24.2	0	0	0	2.2	1.3	4.5	7.2	7.0	4.3	78
Lodge Report	3.1	4.1	6.5	0	0	0	0	0	0	0	0	0	14
AUGUST RETAINED	14.5	19.6	30.7	0	0	0	2.2	1.3	4.5	7.2	7.0	4.3	91
SE	8.3	11.4	12.3	0	0	0	1.2	0.0	3.3	1.3	5.2	3.1	20
TOTAL													
Creel Estimate	65	15	24	0	0	7	2	1	15	7	7	4	149
Lodge Report	17	4	6	0	0	0	0	0	0	0	0	0	28
TOTAL RETAINED	82	20	31	0	0	7	2	1	15	7	7	4	177
SE	40	11	12	0	0	4	1	0	6	1	5	3	45

Estimated numbers of Chum Salmon released by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	0	0	0	0	26.0	0	0	0	26
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	0	0	0	0	26.0	0	0	0	26
SE	0	0	0	0	0	0	0	0	16.3	0	0	0	16
August													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL													
Creel Estimate	0	0	0	0	0	0	0	0	26	0	0	0	26
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	0	0	0	0	0	0	0	0	26	0	0	0	26
SE	0	0	0	0	0	0	0	0	16	0	0	0	16

Estimated numbers of Sockeye Salmon retained by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
August													
Creel Estimate	0	7.7	24.2	0	0	0	0	0	0	0	0	0	32
Lodge Report	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	0	7.7	24.2	0	0	0	0	0	0	0	0	0	32
SE	0	5.7	12.3	0	0	0	0	0	0	0	0	0	14
TOTAL													
Creel Estimate	0	8	24	0	0	0	0	0	0	0	0	0	32
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	0	8	24	0	0	0	0	0	0	0	0	0	32
SE	0	6	12	0	0	0	0	0	0	0	0	0	14

Estimated numbers of Sockeye Salmon released by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
August													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0

Estimated numbers of unknown salmon retained by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
August													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0

Estimated numbers of unknown salmon released by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
August													
Creel Estimate	11.4	0	0	0	0	0	1.1	0.4	0	0	0	0	13
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	11.4	0	0	0	0	0	1.1	0.4	0	0	0	0	13
SE	8.3	0	0	0	0	0	0.9	0.0	0	0	0	0	8
TOTAL													
Creel Estimate	11	0	0	0	0	0	1	0	0	0	0	0	13
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	11	0	0	0	0	0	1	0	0	0	0	0	13
SE	8	0	0	0	0	0	1	0	0	0	0	0	8

Estimated numbers of Pacific Halibut retained by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	414.1	635.9	77.4	0	25.3	396.4	82.4	98.2	488.8	266.9	380.6	347.7	3,214
Lodge Report	70.9	108.9	13.3	0	0	0	0	0	0	0	0	0	193
JUNE RETAINED	485.0	744.7	90.7	0	25.3	396.4	82.4	98.2	488.8	266.9	380.6	347.7	3,407
SE	60.5	78.7	21.0	0	8.8	38.5	25.3	15.7	32.0	58.1	27.1	30.4	137
July													
Creel Estimate	563.6	508.4	181.9	0	33.1	608.9	56.5	144.9	893.1	193.8	432.8	465.9	4,083
Lodge Report	80.9	73.0	26.1	0	0	0	0	0	0	0	0	0	180
JULY RETAINED	644.5	581.4	208.0	0	33.1	608.9	56.5	144.9	893.1	193.8	432.8	465.9	4,263
SE	114.0	88.8	71.2	0	8.4	79.6	13.9	39.5	64.2	49.8	39.2	61.3	214
August													
Creel Estimate	274.3	580.0	278.9	0	49.5	513.4	40.9	83.9	677.8	202.1	133.8	319.4	3,154
Lodge Report	7.0	14.8	7.1	0	0	0	0	0	0	0	0	0	29
AUGUST RETAINED	281.3	594.9	286.0	0	49.5	513.4	40.9	83.9	677.8	202.1	133.8	319.4	3,183
SE	48.1	75.2	58.9	0	11.2	64.4	7.7	1.2	58.9	4.6	25.1	33.7	145
TOTAL													
Creel Estimate	1,252	1,724	538	0	108	1,519	180	327	2,060	663	947	1,133	10,451
Lodge Report	159	197	47	0	0	0	0	0	0	0	0	0	402
TOTAL RETAINED	1,411	1,921	585	0	108	1,519	180	327	2,060	663	947	1,133	10,853
SE	138	140	95	0	17	109	30	42	93	77	54	76	293

Estimated numbers of Pacific Halibut released by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	223.4	381.3	51.9	0	23.8	278.7	50.7	48.9	303.5	147.3	352.0	484.1	2,345
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	223.4	381.3	51.9	0	23.8	278.7	50.7	48.9	303.5	147.3	352.0	484.1	2,345
SE	42.2	76.5	23.8	0	9.1	49.2	23.4	13.1	39.2	35.8	59.8	69.2	150
July													
Creel Estimate	563.4	579.7	147.2	0	15.4	396.1	27.5	117.3	452.4	90.5	377.2	623.0	3,390
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	563.4	579.7	147.2	0	15.4	396.1	27.5	117.3	452.4	90.5	377.2	623.0	3,390
SE	243.2	201.0	55.8	0	5.6	75.0	12.1	57.0	60.5	22.6	63.9	118.1	366
August													
Creel Estimate	297.1	618.7	363.7	0	8.7	168.8	19.9	42.2	343.4	178.9	49.3	366.3	2,457
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	297.1	618.7	363.7	0	8.7	168.8	19.9	42.2	343.4	178.9	49.3	366.3	2,457
SE	100.2	165.7	177.5	0	5.0	45.8	6.3	0.6	51.9	4.0	26.8	66.2	281
TOTAL													
Creel Estimate	1,084	1,580	563	0	48	844	98	208	1,099	417	778	1,473	8,192
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	1,084	1,580	563	0	48	844	98	208	1,099	417	778	1,473	8,192
SE	266	272	188	0	12	101	27	58	89	43	92	152	485

Estimated numbers of Lingcod retained by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	101.6	392.1	68.8	0	2.8	24.0	68.9	53.7	251.1	98.9	136.1	165.4	1,364
Lodge Report	70.5	271.8	47.7	0	0	0	0	0	0	0	0	0	390
JUNE RETAINED	172.1	663.9	116.6	0	2.8	24.0	68.9	53.7	251.1	98.9	136.1	165.4	1,754
SE	9.3	86.7	44.0	0	2.6	9.4	27.8	24.6	37.0	28.6	28.1	34.9	123
July													
Creel Estimate	4.5	509.8	5.0	0	2.2	86.3	8.1	94.6	412.8	55.1	76.4	128.7	1,383
Lodge Report	3.1	350.5	3.4	0	0	0	0	0	0	0	0	0	357
JULY RETAINED	7.6	860.3	8.4	0	2.2	86.3	8.1	94.6	412.8	55.1	76.4	128.7	1,740
SE	4.0	112.6	2.8	0	2.0	29.0	3.8	43.4	59.8	16.8	17.6	26.2	142
August													
Creel Estimate	11.4	309.4	12.1	5.5	0	14.1	5.5	20.6	185.3	57.2	35.2	93.7	750
Lodge Report	0.9	24.2	0.9	0	0	0	0	0	0	0	0	0	26
AUGUST RETAINED	12.3	333.5	13.1	5.5	0	14.1	5.5	20.6	185.3	57.2	35.2	93.7	776
SE	8.3	59.4	8.8	3.7	0	7.0	2.3	0.3	42.0	1.3	19.7	21.2	80
TOTAL													
Creel Estimate	118	1,211	86	5	5	124	83	169	849	211	248	388	3,497
Lodge Report	74	646	52	0	0	0	0	0	0	0	0	0	773
TOTAL RETAINED	192	1,858	138	5	5	124	83	169	849	211	248	388	4,270
SE	13	154	45	4	3	31	28	50	82	33	39	48	205

Estimated numbers of Lingcod released by lodge and non-lodge anglers, by month, 2018.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	24.7	102.3	0	0	5.6	11.6	5.3	59.6	37.9	57.5	26.0	25.3	356
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	24.7	102.3	0	0	5.6	11.6	5.3	59.6	37.9	57.5	26.0	25.3	356
SE	11.2	44.4	0	0	4.0	4.8	4.0	32.0	13.7	22.7	8.6	15.2	65
July													
Creel Estimate	13.4	130.9	0	0	6.0	28.8	36.4	51.9	20.1	29.7	19.2	11.7	348
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	13.4	130.9	0	0	6.0	28.8	36.4	51.9	20.1	29.7	19.2	11.7	348
SE	8.8	48.9	0	0	3.9	21.5	24.1	24.9	8.4	13.2	6.5	8.3	67
August													
Creel Estimate	0	23.2	0	0	0	0	1.1	2.2	18.1	8.9	0	21.3	75
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	23.2	0	0	0	0	1.1	2.2	18.1	8.9	0	21.3	75
SE	0	12.7	0	0	0	0	0.9	0.0	6.4	0.2	0	15.6	21
TOTAL													
Creel Estimate	38	256	0	0	12	40	43	114	76	96	45	58	779
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	38	256	0	0	12	40	43	114	76	96	45	58	779
SE	14	67	0	0	6	22	24	41	17	26	11	23	96

Estimated numbers of other groundfish retained by lodge and non-lodge anglers, by month, 2018. Included are Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and other miscellaneous groundfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	62.2	36.9	26.7	0	11.1	78.8	8.8	3.0	83.6	13.8	90.2	10.0	425
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	62.2	36.9	26.7	0	11.1	78.8	8.8	3.0	83.6	13.8	90.2	10.0	425
SE	16.6	18.8	16.6	0	7.9	25.6	5.1	0.1	20.9	9.2	29.7	6.7	56
July													
Creel Estimate	31.4	53.3	2.5	23.9	7.9	61.1	4.4	0	276.0	7.1	82.7	6.2	557
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	31.4	53.3	2.5	23.9	7.9	61.1	4.4	0	276.0	7.1	82.7	6.2	557
SE	13.1	25.8	2.0	15.8	3.5	18.7	2.8	0	63.9	7.1	25.9	4.2	79
August													
Creel Estimate	34.3	15.5	0	10.9	14.6	35.2	11.1	11.2	67.8	1.8	3.5	0	206
Lodge Report	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	34.3	15.5	0	10.9	14.6	35.2	11.1	11.2	67.8	1.8	3.5	0	206
SE	18.5	11.4	0	9.0	6.7	13.1	5.1	0.1	19.7	0.0	2.6	0	35
TOTAL													
Creel Estimate	128	106	29	35	34	175	24	14	427	23	176	16	1,187
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	128	106	29	35	34	175	24	14	427	23	176	16	1,187
SE	28	34	17	18	11	34	8	0	70	12	39	8	103

Estimated numbers of other groundfish released by lodge and non-lodge anglers, by month, 2018. Included are Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and other miscellaneous groundfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	71.7	94.9	60.8	0	53.3	237.7	159.6	6.4	66.1	0	17.5	18.4	786
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	71.7	94.9	60.8	0	53.3	237.7	159.6	6.4	66.1	0	17.5	18.4	786
SE	22.3	45.7	22.9	0	17.9	73.2	119.9	0.2	41.4	0	8.3	13.7	159
July													
Creel Estimate	17.9	52.4	32.2	41.4	107.3	190.7	2.6	0	217.0	0	22.5	3.9	688
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	17.9	52.4	32.2	41.4	107.3	190.7	2.6	0	217.0	0	22.5	3.9	688
SE	9.7	29.3	15.3	23.4	33.0	50.6	1.6	0	64.4	0	9.4	3.9	98
August													
Creel Estimate	114.3	108.3	97.0	20.8	64.0	133.6	37.6	35.0	198.8	109.1	197.1	21.3	1,137
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	114.3	108.3	97.0	20.8	64.0	133.6	37.6	35.0	198.8	109.1	197.1	21.3	1,137
SE	68.4	43.5	39.0	8.9	20.0	35.4	12.2	0.2	48.2	1.1	67.3	10.4	130
TOTAL													
Creel Estimate	204	256	190	62	225	562	200	41	482	109	237	44	2,611
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	204	256	190	62	225	562	200	41	482	109	237	44	2,611
SE	73	70	48	25	43	96	121	0	91	1	68	18	227

Estimated numbers of rockfish retained by lodge and non-lodge anglers, by month, 2018. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	153.3	301.7	87.1	2.3	28.2	262.0	202.7	84.2	434.6	184.1	309.5	290.3	2,340
Lodge Report	90.2	151.6	57.2	0	0	0	0	0	0	0	0	0	299
JUNE RETAINED	243.4	453.3	144.4	2.3	28.2	262.0	202.7	84.2	434.6	184.1	309.5	290.3	2,639
SE	50.6	55.5	41.5	1.8	12.8	50.6	44.1	26.2	48.3	49.0	49.6	57.5	152
July													
Creel Estimate	89.6	849.5	150.2	1.0	87.3	518.2	57.3	149.1	634.2	96.2	255.4	178.8	3,067
Lodge Report	23.6	249.0	48.4	0	0	0	0	0	0	0	0	0	321
JULY RETAINED	113.2	1098.6	198.6	1.0	87.3	518.2	57.3	149.1	634.2	96.2	255.4	178.8	3,388
SE	39.8	238.5	53.7	0.9	22.3	90.8	15.3	57.9	73.5	31.2	51.9	38.3	290
August													
Creel Estimate	57.1	402.2	194.0	5.5	2.9	203.9	42.0	59.7	429.3	139.5	35.2	268.3	1,840
Lodge Report	3.7	19.2	13.1	0	0	0	0	0	0	0	0	0	36
AUGUST RETAINED	60.8	421.4	207.1	5.5	2.9	203.9	42.0	59.7	429.3	139.5	35.2	268.3	1,876
SE	27.5	73.7	67.1	3.7	2.2	42.3	9.1	0.4	64.8	4.5	10.1	71.7	148
TOTAL													
Creel Estimate	300	1,553	431	9	118	984	302	293	1,498	420	600	737	7,247
Lodge Report	117	420	119	0	0	0	0	0	0	0	0	0	656
TOTAL RETAINED	417	1,973	550	9	118	984	302	293	1,498	420	600	737	7,903
SE	70	256	95	4	26	112	48	64	109	58	73	100	360

Estimated numbers of rockfish released by lodge and non-lodge anglers, by month, 2018. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	192.7	467.1	154.1	0	212.8	365.9	519.0	161.0	490.0	437.2	568.2	710.8	4,279
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	192.7	467.1	154.1	0	212.8	365.9	519.0	161.0	490.0	437.2	568.2	710.8	4,279
SE	67.5	94.1	45.8	0	45.9	61.2	126.6	41.5	52.4	97.1	89.0	178.3	302
July													
Creel Estimate	116.4	1009.5	280.5	6.4	369.2	1300.5	244.0	691.9	1186.5	243.3	483.3	493.7	6,425
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	116.4	1009.5	280.5	6.4	369.2	1300.5	244.0	691.9	1186.5	243.3	483.3	493.7	6,425
SE	48.7	213.1	105.3	3.5	103.7	374.1	62.0	185.5	234.1	60.4	77.5	112.1	570
August													
Creel Estimate	217.1	835.2	254.6	10.9	43.7	717.3	55.3	88.9	659.7	121.6	45.8	221.5	3,272
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	217.1	835.2	254.6	10.9	43.7	717.3	55.3	88.9	659.7	121.6	45.8	221.5	3,272
SE	73.0	249.8	78.7	9.0	13.4	138.1	13.2	1.0	109.1	2.5	16.0	52.6	329
TOTAL													
Creel Estimate	526	2,312	689	17	626	2,384	818	942	2,336	802	1,097	1,426	13,975
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	526	2,312	689	17	626	2,384	818	942	2,336	802	1,097	1,426	13,975
SE	111	342	139	10	114	403	142	190	264	114	119	217	724

APPENDIX 5: ADIPOSE FIN-CLIP INCIDENCE

Incidence of fin clipping observed in biosamples, by species, month and subarea, in 2018.

Species		Subarea														TOTAL
Month	Adipose	3I	3J	3K	3?	4A	4B	4C	4D	4E	4F	4G	4H	4L	4?	
Chinook Salmon																
June	Clipped	0	2	2	0	1	0	2	0	0	1	0	0	0	0	8
	Not Clipped	0	8	6	0	2	8	14	4	0	22	3	7	0	0	74
	Percent Clipped	-	20%	25%	-	33%	0%	13%	0%	-	4%	0%	0%	-	-	10%
July	Clipped	0	0	2	0	0	0	1	1	0	12	0	2	2	0	20
	Not Clipped	1	3	3	0	0	6	8	6	2	41	2	2	4	0	78
	Percent Clipped	0%	0%	40%	-	-	0%	11%	14%	0%	23%	0%	50%	33%	-	20%
August	Clipped	3	1	0	0	0	2	0	0	0	1	0	1	0	0	8
	Not Clipped	9	4	4	0	0	2	4	6	0	10	0	5	1	0	45
	Percent Clipped	25%	20%	0%	-	-	50%	0%	0%	-	9%	-	17%	0%	-	15%
Total	Clipped	3	3	4	0	1	2	3	1	0	14	0	3	2	0	36
	Not Clipped	10	15	13	0	2	16	26	16	2	73	5	14	5	0	197
	Percent Clipped	23%	17%	24%	-	33%	11%	10%	6%	0%	16%	0%	18%	29%	-	15%
Coho Salmon																
June	Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Not Clipped	0	0	0	0	0	0	0	0	3	2	0	0	0	0	5
	Percent Clipped	-	-	-	-	-	-	-	-	0%	0%	-	-	-	-	0.0%
July	Clipped	0	0	0	0	0	0	0	0	3	1	0	1	0	0	5
	Not Clipped	3	2	2	0	0	3	6	4	15	109	1	8	41	0	194
	Percent Clipped	0%	0%	0%	-	-	0%	0%	0%	17%	1%	0%	11%	0%	-	2.5%
August	Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Not Clipped	10	10	27	0	0	5	2	12	0	19	0	13	3	1	102
	Percent Clipped	0%	0%	0%	-	-	0%	0%	0%	-	0%	-	0%	0%	0%	0.0%
Total	Clipped	0	0	0	0	0	0	0	0	3	1	0	1	0	0	5
	Not Clipped	13	12	29	0	0	8	8	16	18	130	1	21	44	1	301
	Percent Clipped	0%	0%	0%	-	-	0%	0%	0%	14%	1%	0%	5%	0%	0%	1.6%